2026 PRELIMS 99% MASTERY · CONSOLIDATED CHAPTER-WISE FACTS

SCIENCE & TECHNOLOGY

सामान्य विज्ञान एवं प्रौद्योगिकी — संपूर्ण उच्च-प्राथमिकता संकलन

Cell Biology & Genetics, Human Physiology & Systems, Vitamins & Diseases, Daily Life Chemistry, Physics Laws & Units, Space, Defence & Nuclear Tech.

UPPCS & UKPCS Prelims 2026 • High-Yield Master Revision Series • 99% Coverage Edition

Science & Technology — Master Chapter Index

Consolidated High-Yield Revision Matrix • All 27 Chapters

Ch 01. 00_Daily_Revisi
00 — Executive Bird's-Eye Master Matrix

Ch 02. 01_Cell_Genetic
Living World, Cell, Genetics and Biotechnology

Ch 03. 02_Animal_Biolo
Animal Biology and Husbandry

Ch 04. 03_Nutrition_Di
Nutrition, Vitamins, Diseases and Medicine

Ch 05. 04_Digestion_Re
Digestion, Respiration and Excretion

Ch 06. 05_Blood_Heart_
Blood, Heart, Circulation and Lymph

Ch 07. 06_Control_Repr
Nervous System, Hormones, Reproduction and Support

Ch 08. 07_Plants_Agric
Plants, Agriculture and Plant Diseases

Ch 09. 08_Microbiology
Microbiology, Environment and Applied Science

Ch 10. 09_Physics_Fund
Physics Fundamentals, Measurement and Instruments

Ch 11. 10_Mechanics_an
Mechanics, Gravitation and Physical Properties of Matter

Ch 12. 11_Energy_Heat_
Energy, Heat and Thermal Science

Ch 13. 12_Light_Optics
Light, Optics and Laser Technology

Ch 14. 13_Sound_and_Wa
Sound and Wave Motion

Ch 15. 14_Electricity_
Electricity, Magnetism and Electromagnetism

Ch 16. 15_Electronics_
Electronics, Semiconductors and Computer Technology

Ch 17. 16_Nuclear_and_
Nuclear and Atomic Physics

Ch 18. 17_Scientists_D
Scientists, Discoveries, Defence & Space Technology, and Applicat

Ch 19. 18_Atomic_Struc
Chemistry Topic 1 — Atomic Structure and Periodic Classification

Ch 20. 19_Matter_Solut
Chemistry Topic 2 — Matter, Solutions and Purification

Ch 21. 20_Metals_and_C
Chemistry Topic 3 — Metals, Non-Metals and Chemical Reactions

Ch 22. 21_Acids_Bases_
Chemistry Topic 4 — Acids, Bases, Salts and Carbohydrates

Ch 23. 22_Carbon_Organ
Chemistry Topic 5 — Carbon, Organic Compounds, Polymers and Fibre

Ch 24. 23_Gases_Fuels_
Chemistry Topic 6 — Gases, Fuels, Energy and Combustion

Ch 25. 24_Chemistry_Da
Chemistry Topic 7 — Chemistry in Daily Life, Industry and Agricul

Ch 26. 25_Radiation_Nu
Chemistry Topic 8 — Nuclear Chemistry, Explosives and Environment

Ch 27. 26_Chemistry_Di
Chemistry Topic 9 — Discoveries, Scientists and Miscellaneous Che

Chapter 01: 00 — Executive Bird's-Eye Master Matrix

★ UPPCS & UKPCS Prelims Daily Fact Dump · Active Recall Cheat-Sheet

Exam Focus: UPPSC Prelims (Paper 1), UKPSC Prelims (Paper 1), RO/ARO, BEO
Frequency: Daily morning read (15–20 mins)
Question Types Covered: Vitamins, Chemical Names & Deficiency Diseases, Pathogens vs Infectious Diseases (Viral, Bacterial, Protozoan), Human Physiology & Endocrine Glands, Plant Hormones, Units & Dimensions, Optics (Lenses, Mirrors, TIR, Rainbow), Sound & Electromagnetic Spectrum, Common Chemicals & Commercial Names, Alloys & Metal Compositions, Everyday pH Values, Indian Space Missions (ISRO) & Defence Tech (DRDO Missiles, Nuclear Power Plants).

1. Complete Vitamin Master Reference & Deficiency Diseases

VITAMIN	CHEMICAL / SCIENTIFIC NAME	FAT / WATER SOLUBLE	KEY DEFICIENCY DISEASE / SYMPTOMS	PRIMARY NATURAL FOOD SOURCES
Vitamin A	**Retinol**	Fat-soluble	**Night blindness (Nyctalopia)**, **Xerophthalmia** (dry cornea), Keratomalacia	Carrot, Papaya, Mango, Liver, Fish oil, Golden Rice
Vitamin B1	**Thiamine**	Water-soluble	**Beri-beri** (Peripheral neuropathy / cardiac failure), Wernicke-Korsakoff syndrome	Unpolished rice, yeast, whole grains (*lost in polished rice*)
Vitamin B2	**Riboflavin**	Water-soluble	**Cheilosis** (cracking at corners of mouth), Glossitis (red swollen tongue)	Milk, eggs, liver (*gives yellow colour to milk/whey*)
Vitamin B3	**Niacin (Nicotinic acid)**	Water-soluble	**Pellagra** (4Ds: Dermatitis, Diarrhea, Dementia, Death)	Yeast, meat, fish, legumes
Vitamin B5	**Pantothenic acid**	Water-soluble	Burning feet syndrome, fatigue	Mushroom, eggs, broccoli
Vitamin B6	**Pyridoxine**	Water-soluble	Peripheral neuropathy, microcytic anemia, convulsions in infants	Cereals, bananas, meat
Vitamin B7	**Biotin (Vitamin H)**	Water-soluble	Hair loss (alopecia), dermatitis	Egg yolk, nuts, intestinal flora (*avidin in raw egg binds biotin*)
Vitamin B9	**Folic Acid (Folate)**	Water-soluble	**Megaloblastic anemia**, Neural tube defects in fetuses (Spina bifida)	Green leafy vegetables, citrus fruits
Vitamin B12	**Cyanocobalamin**	Water-soluble	**Pernicious anemia**, demyelination of spinal cord (*contains Cobalt metal*)	Meat, fish, eggs, dairy (*Absent in plant foods*)
Vitamin C	**Ascorbic acid**	Water-soluble	**Scurvy** (bleeding gums, petechiae, impaired wound healing); heat-sensitive (*destroyed by cooking*)	Aonla (highest source), Guava, Citrus fruits (Lemon, Orange)
Vitamin D	**Calciferol** (D_2: Ergocalciferol, D_3: Cholecalciferol)	Fat-soluble	**Rickets** in children (bowed legs), **Osteomalacia** in adults (*hormone-like vitamin*)	Sunlight synthesis in skin, Cod liver oil
Vitamin E	**Tocopherol**	Fat-soluble	Infertility / reproductive failure, muscle weakness, hemolytic anemia (*Beauty Vitamin / Anti-oxidant*)	Wheat germ oil (richest), sunflower seeds, almonds
Vitamin K	**Phylloquinone** (K_1), Menaquinone (K_2)	Fat-soluble	**Defective blood clotting**, prolonged bleeding time (needed for synthesis of prothrombin in liver)	Spinach, kale, synthesized by human gut bacteria

Fat-Soluble vs Water-Soluble Mnemonic:
Fat-soluble: **K - E - D - A** (stored in liver and adipose tissue, hypervitaminosis possible).
Water-soluble: **B-complex and C** (excreted in urine, regular intake required).

2. Infectious Diseases & Causative Pathogens Master Matrix

A. Viral Diseases

DISEASE	CAUSATIVE VIRUS	MODE OF TRANSMISSION / KEY DIAGNOSTIC FEATURE
COVID-19	SARS-CoV-2 (enveloped positive-sense RNA)	Respiratory droplets; RT-PCR test
Rabies (Hydrophobia)	Lyssavirus / Rhabdovirus (bullet-shaped RNA)	Bite of rabid dogs, bats; affects central nervous system
Poliomyelitis	Poliovirus (Enterovirus, single-stranded RNA)	Fecal-oral route; destroys anterior horn motor neurons of spinal cord; Oral Polio Vaccine (Sabin), Inactivated Polio Vaccine (Salk)
AIDS	Human Immunodeficiency Virus (HIV - Retrovirus with Reverse Transcriptase)	Blood, sexual, maternal; attacks **CD4+ Helper T-cells**; ELISA test, Western Blot confirmatory
Dengue (Break-bone fever)	Flavivirus (DEN-1 to DEN-4, ssRNA)	Transmitted by **Aedes aegypti** mosquito (day-biting); severe thrombocytopenia (platelet drop)
Chikungunya	Chikungunya virus (Alphavirus)	Transmitted by **Aedes aegypti** and *Aedes albopictus*; severe joint pain
Yellow Fever	Flavivirus	*Aedes aegypti* mosquito vector
Zika Virus	Flavivirus	*Aedes* mosquito; causes **Microcephaly** in newborns
Hepatitis B	Hepatitis B Virus (HBV - **DNA virus**, whereas Hep A, C, D, E are RNA)	Blood, body fluids; attacks liver; vaccine available
Smallpox	Variola virus (DNA virus)	Completely **eradicated from world in 1980** (WHO)
Measles, Mumps, Rubella	Morbillivirus, Paramyxovirus, Rubivirus	MMR vaccine

B. Bacterial Diseases

DISEASE	CAUSATIVE BACTERIUM	KEY CLINICAL FEATURE / DIAGNOSTIC TEST
Tuberculosis (TB)	*Mycobacterium tuberculosis* (acid-fast bacillus)	Affects lungs; **Mantoux test**; **BCG vaccine** (Bacillus Calmette-Guérin)
Cholera	*Vibrio cholerae* (comma-shaped)	"*Rice water stools*", massive dehydration; fecal-oral route
Typhoid (Enteric fever)	*Salmonella typhi*	Step-ladder fever pattern, intestinal perforation; **Widal test**
Tetanus (Lockjaw)	*Clostridium tetani* (anaerobic, drumstick-shaped)	Neurotoxin tetanospasmin causes severe muscle spasms; ATS / Tetanus toxoid
Diphtheria	*Corynebacterium diphtheriae*	Grey pseudomembrane over throat / larynx; **Schick test**
Pertussis (Whooping Cough)	*Bordetella pertussis*	Paroxysmal coughing ending in whoop; DPT triple vaccine
Plague (Black Death)	*Yersinia pestis*	Transmitted by **Xenopsylla cheopis** (oriental rat flea); Bubonic, Pneumonic forms
Leprosy (Hansen's Disease)	*Mycobacterium leprae*	Hypopigmented skin patches with loss of sensation; Lepromin test
Anthrax	*Bacillus anthracis* (spore-forming)	Zoonotic disease of cattle; potential bioweapon
Syphilis & Gonorrhea	*Treponema pallidum* & *Neisseria gonorrhoeae*	Sexually transmitted bacterial infections

C. Protozoan & Other Pathogen Diseases

- **Malaria:** Caused by protozoan **Plasmodium** (*P. vivax, P. falciparum* [deadliest malignant tertian], *P. malariae, P. ovale*). Vector: **Female Anopheles mosquito**. Infective stage to humans: **Sporozoite**. Parasite multiplies in liver cells and destroys red blood cells (**RBCs**), releasing **hemozoin toxin** causing periodic shivering chills and high fever. Discovered by **Sir Ronald Ross** (in Secunderabad, India - Nobel Prize 1902). Treatment: Quinine (extracted from **Cinchona bark**), Artemisinin.
- **Kala-Azar (Visceral Leishmaniasis / Dum-Dum Fever):** Caused by *Leishmania donovani*. Vector: **Phlebotomus sandfly**. Enlarges spleen and liver (hepatosplenomegaly).
- **Sleeping Sickness (African Trypanosomiasis):** Caused by *Trypanosoma brucei*. Vector: **Tsetse fly** (*Glossina*).
- **Amoebic Dysentery:** Caused by *Entamoeba histolytica*.
- **Ringworm (Tinea):** Fungal infection (*Trichophyton, Microsporum, Epidermophyton*).
- **Filariasis / Elephantiasis:** Caused by parasitic nematode worm *Wuchereria bancrofti*. Transmitted by **Culex mosquito**. Blocks lymphatic vessels causing swelling of limbs/scrotum.

3. Human Physiology & Endocrine Hormones

GLAND	HORMONES PRODUCED	PRIMARY FUNCTION & PRELIMS EXAM TRIGGERS
Pituitary (Master Gland)	Growth Hormone (GH), TSH, ACTH, FSH, LH, Prolactin, **Oxytocin**, **Vasopressin (ADH)**	Regulated by Hypothalamus. Hypersecretion of GH causes **Gigantism / Acromegaly**; hyposecretion causes **Dwarfism**. **ADH (Antidiuretic Hormone)** controls water reabsorption in kidneys; deficiency causes **Diabetes Insipidus** (frequent urination without sugar). **Oxytocin** is the milk-ejection & birth hormone.
Thyroid	**Thyroxine (T_4)**, Triiodothyronine (T_3), Calcitonin	Largest endocrine gland; requires **Iodine**. Deficiency causes **Simple Goitre** (enlarged neck), **Cretinism** in infants, **Myxedema** in adults. Hyperthyroidism causes **Grave's Disease** (exophthalmic goitre).
Parathyroid	Parathyroid Hormone (PTH / Collip's hormone)	**Increases blood calcium levels** (hypercalcemic) by dissolving bone; acts antagonistically to thyroid calcitonin.
Pancreas (Islets of Langerhans)	**Insulin** (from β-cells), **Glucagon** (from α-cells), Somatostatin (from δ-cells)	**Insulin lowers blood glucose** by converting glucose to glycogen (glycogenesis). Deficiency causes **Diabetes Mellitus** (hyperglycemia, glycosuria). Discovered by Banting & Best (1921). Glucagon raises blood glucose.
Adrenal Gland	Adrenaline (Epinephrine) & Noradrenaline; Cortisol, Aldosterone	Adrenaline is the **"Emergency Hormone" / "Fight, Flight or Fright" (3F/4S hormone)**; increases heart rate, breathing, blood pressure.
Pineal Gland	**Melatonin**	Regulates circadian rhythm (24-hour sleep-wake cycle).

Blood Groups & ABO System: Discovered by **Karl Landsteiner** (1900).
- **Universal Donor: O^- (O Negative)** — has neither A nor B antigens and no Rh factor on RBCs.
- **Universal Recipient: AB^+ (AB Positive)** — has both A and B antigens and Rh factor; has no anti-A or anti-B antibodies in plasma.
- **Normal Blood Pressure: 120/80 mm Hg** (Systolic/Diastolic), measured by **Sphygmomanometer**. Blood pH: **7.35–7.45 (slightly basic)**.

4. Optics & Physics Principles (High-Frequency Exam Traps)

PHENOMENON	UNDERLYING OPTICAL PRINCIPLE	EXAM APPLICATION / EXAMPLES
Mirage in deserts / Looming in cold	**Total Internal Reflection (TIR)**	Optical illusion; light travels from denser to rarer medium at angle $>$ critical angle.
Sparkling of Diamond	**Total Internal Reflection (TIR)**	Very small critical angle of diamond ($\approx 24.4^{\circ}$) due to high refractive index (≈ 2.42).
Endoscope / Optical Fibre Cables	**Total Internal Reflection (TIR)**	Zero signal loss communication; invented by Narinder Singh Kapany.
Blue colour of sky / Red colour of sunset	**Rayleigh Scattering of Light**	Scattering $\propto \frac{1}{\lambda^4}$. Blue light (shorter wavelength) scatters most; red light (longest wavelength) scatters least and reaches eyes at horizon.
Twinkling of Stars	**Atmospheric Refraction**	Continuously fluctuating refractive index of atmospheric air layers.
Coin appears raised / Stick bent in water	**Refraction of Light**	Light bends away from normal when entering air from water.
Rainbow Formation	**Dispersion $+$ Refraction $+$ Total Internal Reflection**	Water droplets act as tiny prisms. Primary rainbow: 2 refractions $+$ 1 internal reflection; Secondary rainbow: 2 refractions $+$ 2 internal reflections.
Soap bubble shining with brilliant colours	**Interference of Light**	Destructive and constructive interference of light reflected from upper and lower thin film surfaces.

Common Eye Defects & Corrective Lenses

1. **Myopia (Short-sightedness):** Nearby objects seen clearly, distant objects blurred. Image formed **in front of retina**. Corrected by **Concave Lens** (diverging lens).
2. **Hypermetropia (Long-sightedness):** Distant objects seen clearly, nearby objects blurred. Image formed **behind retina**. Corrected by **Convex Lens** (converging lens).
3. **Presbyopia (Old age vision):** Loss of accommodation power of ciliary muscles. Corrected by **Bifocal Lens** (upper part concave for distance, lower part convex for reading).
4. **Astigmatism:** Inability to focus simultaneously on horizontal and vertical lines due to irregular cornea curvature. Corrected by **Cylindrical Lens**.

5. Commercial Names of Common Chemical Compounds

CHEMICAL / COMMON COMMERCIAL NAME	CHEMICAL FORMULA	SCIENTIFIC / IUPAC NAME	HIGH-YIELD APPLICATION
Baking Soda	NaHCO_3	Sodium hydrogen carbonate	Baking, antacid, fire extinguisher
Washing Soda	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	Sodium carbonate decahydrate	Glass, soap, water softening
Caustic Soda	NaOH	Sodium hydroxide	Soap manufacturing, petroleum refining
Caustic Potash	KOH	Potassium hydroxide	Soft soaps, alkaline batteries
Bleaching Powder	CaOCl_2	Calcium hypochlorite	Disinfectant for water, bleaching cotton
Quicklime	CaO	Calcium oxide	Cement manufacture, white washing
Slaked Lime	Ca(OH)_2	Calcium hydroxide	Mortar, neutralizing soil acidity
Plaster of Paris (POP)	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$	Calcium sulphate hemihydrate	Bone fractures, statues, false ceilings
Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Calcium sulphate dihydrate	Retarder in cement setting, soil reclamation
Laughing Gas	N_2O	Nitrous oxide	Dental anaesthetic, propellant
Tear Gas	CCl_3NO_2 (Chloropicrin) / CS gas	2-chlorobenzalmalononitrile	Riot control
Heavy Water	D_2O	Deuterium oxide	Moderator in nuclear reactors
Dry Ice	Solid CO_2	Solid carbon dioxide	Refrigerant, sublimes without melting
Epsom Salt	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	Magnesium sulphate heptahydrate	Laxative, bath salt
Blue Vitriol	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	Copper sulphate pentahydrate	Fungicide (Bordeaux mixture), electroplating
Green Vitriol	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	Ferrous sulphate heptahydrate	Ink manufacturing
White Vitriol	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	Zinc sulphate heptahydrate	Eye drops, pigments

6. Major Alloys & Their Compositions

ALLOY	MAJOR CONSTITUENTS (%)	PRIMARY PROPERTIES & APPLICATIONS
Brass (Pital)	**Copper (60–70%) $+$ Zinc (30–40%)**	Utensils, musical instruments, cartridge cases (*No Tin*)
Bronze (Kansa)	**Copper (88%) $+$ Tin (12%)**	Statues, medals, coins, bells (*No Zinc*)
German Silver	**Copper (50%) $+$ Zinc (30%) $+$ Nickel (20%)**	**Contains ZERO Silver!** Utensils, artificial jewellery
Stainless Steel	**Iron $+$ Chromium (18%) $+$ Nickel (8%) $+$ Carbon (0.1–1%)**	Rust-resistant; Chromium gives corrosion resistance
Solder (Tanka)	**Lead (50%) $+$ Tin (50%)**	Low melting point; joining electrical wires
Duralumin	**Aluminium (95%) $+$ Copper (4%) $+$ Magnesium (0.5%) $+$ Manganese (0.5%)**	Light and strong; aircraft bodies, pressure cookers
Gun Metal	**Copper (88%) $+$ Tin (10%) $+$ Zinc (2%)**	Heavy machinery gears, gun barrels
Bell Metal	**Copper (78%) $+$ Tin (22%)**	High resonance for casting bells
Amalgam	**Mercury $+$ Any other metal** (except Iron, Platinum, Tungsten)	Dental fillings (Silver-Mercury amalgam); **Iron does not form amalgam**
Nichrome	**Nickel (60%) $+$ Chromium (40%)**	High resistance & high melting point; electric heater coils

7. Indian Space Programme (ISRO) & Defence Tech

- **ISRO Founded: 15 August 1969**; Headquarters: **Bengaluru**; Space Commission & Department of Space (DoS) formed in 1972.
- **Launch Port:** Satish Dhawan Space Centre (SDSC - SHAR), **Sriharikota (Andhra Pradesh)**, on Pulicat Lake barrier island. Second spaceport under construction at **Kulasekarapattinam (Tamil Nadu)** for SSLV launches.
- **Key Historic Missions:**
- **Aryabhata:** India's first satellite, launched on **19 April 1975** by Soviet Kosmos-3M rocket.
- **Chandrayaan-1:** Launched on **22 October 2008** via PSLV-C11; Moon Impact Probe detected water molecules (H_2O/OH) on lunar surface.
- **Mangalyaan (Mars Orbiter Mission - MOM):** Launched on **5 November 2013** via PSLV-C25; inserted into Martian orbit on **24 September 2014** on maiden attempt.
- **Chandrayaan-3:** Launched on **14 July 2023** via LVM3-M4; successfully achieved historic **soft landing near lunar South Pole** on **23 August 2023** (celebrated as **National Space Day**). Lander: *Vikram*, Rover: *Pragyan*. Landing site named **Shiv Shakti Point**.
- **Aditya-L1:** India's first solar observatory launched on **2 September 2023** via PSLV-C57; inserted into halo orbit around **Lagrange Point 1 (L1)** on 6 January 2024.
- **DRDO Missiles (Integrated Guided Missile Development Programme - IGMDP, 1983 - Dr. APJ Abdul Kalam):**
- **P - Prithvi:** Surface-to-Surface short-range ballistic missile (Liquid fuel).
- **A - Agni:** Surface-to-Surface medium-to-intercontinental ballistic missile (Agni I to V, Solid fuel, MIRV tech).
- **T - Trishul:** Surface-to-Air short-range missile.
- **N - Nag:** Anti-Tank guided missile ("*Fire and Forget*"); helicopter-launched version is **Helina / Dhruvastra**.
- **A - Akash:** Surface-to-Air medium-range missile with Rajendra multi-target phased array radar.
- **Nuclear Power Plants in India:** 1. **Narora Atomic Power Station (NAPS): Bulandshahr, Uttar Pradesh** (PHWR, only nuclear plant in UP) 2. **Tarapur:** Maharashtra (Oldest nuclear plant in India, 1969) 3. **Rawatbhata:** Rajasthan (near Kota, Chambal river) 4. **Kakrapar:** Gujarat (Tapi river) 5. **Kaiga:** Karnataka (Kali river) 6. **Kalpakkam (MAPS):** Tamil Nadu (Indira Gandhi Centre for

Atomic Research) 7. **Kudankulam:** Tamil Nadu (Largest nuclear power capacity in India, VVER Russian reactors) 8. **Gorakhpur Nuclear Plant:** Under construction in **Fatehabad, Haryana** (Not in Gorakhpur, UP!)

Chapter 02: Living World, Cell, Genetics and Biotechnology

🎯 Consolidated Must-Score Facts

1. **Key Specialized Biological Branches**:
2. **Key Agricultural & Applied Culture Terminologies**:
3. **Origin of Life (Oparin-Haldane Hypothesis)**:
4. **The Primitive Earth Atmosphere**:
5. **The Miller-Urey Experiment (1953)**:
6. **Geological Chronology of Cenozoic Epochs (Oldest to Most Recent)**:
7. **Lamarckism (Theory of Inheritance of Acquired Characteristics)**:
8. **Darwinism (Theory of Natural Selection)**:
9. **Neo-Darwinism & Modern Synthetic Theory**:
10. **Hugo de Vries' Mutation Theory (1901)**:
11. **Landmark Evolutionary Fossils & Missing Links**:
12. **Augustin Pyramus de Candolle (A. P. de Candolle)**: Swiss botanist who **first coined the term 'Taxonomy'** and was the **first to use vascular tissue characteristics** to classify plants into *Vasculares* (with vascular bundles) and *Cellulares* (without vascular bundles).
13. **Carolus Linnaeus** (*Systema Naturae*, 1758) established the **Binomial Nomenclature** and is the Father of Modern Taxonomy; scientific names consist of **Genus** (capitalised) and **species epithet** (lowercase), printed in *italics*.
14. **R. H. Whittaker (1969)** proposed the **Five-Kingdom System**: **Monera** (prokaryotes), **Protista** (unicellular eukaryotes), **Fungi** (chitinous heterotrophs), **Plantae** (autotrophs with cellulose wall), and **Animalia** (heterotrophs without cell wall).
15. **Carl Woese (1990)** introduced the **Three-Domain System** based on 16S rRNA sequences: **Bacteria**, **Archaea**, and **Eukarya**; Archaebacteria have branched ether lipids allowing them to survive in extreme environments (methanogens in ruminant guts, halophiles in salt lakes, thermoacidophiles in hot springs).
16. **Antony van Leeuwenhoek (1632–1723)** is the Father of Microbiology; discovered bacteria ("animalcules") using single high-power lenses; bacteria are 0.5–5.0 μm in size and visible under a compound microscope.
17. **Prokaryotes** (bacteria, cyanobacteria) have naked circular DNA in a **nucleoid**, lack membrane-bound organelles, and possess **70S ribosomes** (50S + 30S); **Eukaryotes** have membrane-bound nuclei, linear DNA with histones, and cytoplasmic **80S ribosomes** (60S + 40S).
18. **Cell Wall** chemical facts: Plants = **Cellulose + Hemicellulose + Pectin**; Fungi = **Chitin** (polymer of N-acetylglucosamine); Bacteria = **Peptidoglycan / Murein**; Algae = **Cellulose + Galactans + Mannans + $CaCO_3$**.
19. **Singer and Nicolson (1972)** proposed the **Fluid Mosaic Model** of the plasma membrane: a quasi-fluid bilayer of phospholipids with protein "icebergs floating in a lipid sea"; selectively permeable.
20. **Mitochondria** (Kolliker 1857, Benda 1898) are the powerhouses of the cell (ATP generation via oxidative phosphorylation on inner membrane cristae); contain circular DNA and 70S ribosomes (semi-autonomous); inherited maternally.
21. **Plastids** (Schimper): **Chloroplasts** (photosynthesis, contain chlorophyll and circular DNA); **Chromoplasts** (pigment storage: Lycopene in tomato, Capsanthin in red chilli, Carotene in carrot, Betanin in beetroot); **Leucoplasts** (storage: Amyloplasts = starch, Elaioplasts = oils/fats, Aleuroplasts = proteins).
22. **Endoplasmic Reticulum (ER)**: **Rough ER (RER)** has 80S ribosomes on its surface and synthesises proteins; **Smooth ER (SER)** lacks ribosomes and synthesises lipids, steroid hormones, and detoxifies drugs in the liver.
23. **Golgi Apparatus** (Camillo Golgi, 1898) packages and secretes macromolecules, modifies proteins (glycosylation to form glycoproteins and glycolipids), and forms the **acrosome** of sperm and primary lysosomes.
24. **Lysosomes** (Christian de Duve, 1955) are "suicidal bags" containing hydrolytic enzymes (acid hydrolases active at pH 4.5–5.0) that digest cellular waste and foreign pathogens.
25. **Ribosomes** (George Palade, 1953) are non-membrane-bound ribonucleoprotein particles (rRNA + protein) responsible for protein synthesis; known as the "protein factories" of the cell.
26. **Centrosome / Centrioles** (Boveri, 1888) have a $9+0$ cartwheel microtubule arrangement; present in animal cells to form the mitotic spindle and basal bodies; **absent in higher plant cells**.

27. **Peroxisomes** contain catalase and urate oxidase to break down toxic H_2O_2 and perform photorespiration in plants; **Glyoxysomes** convert fats into carbohydrates via the glyoxylate cycle in germinating fatty seeds (castor, groundnut).
28. **Nucleus** was discovered by **Robert Brown (1831)**; the **Nucleolus** (Fontana 1781) is the site of active ribosomal RNA (**rRNA**) synthesis; **Chromatin** was named by Walther Flemming.
29. **Watson and Crick (1953)** proposed the **double-helix model of DNA** based on Rosalind Franklin and Maurice Wilkins' X-ray crystallography; won the Nobel Prize in 1962.
30. **Chargaff's Rule**: In double-stranded DNA, the ratio of Adenine to Thymine and Guanine to Cytosine is equal, i.e., $A + G = T + C$ or $(A+G)/(T+C) = 1$.
31. **RNA** is typically single-stranded, contains **Ribose sugar**, and replaces Thymine with **Uracil**; three functional types: mRNA (messenger), tRNA (transfer / soluble adaptor with cloverleaf secondary structure), and rRNA (ribosomal, structural and catalytic).
32. **Mitosis** (equational division) occurs in somatic cells, yielding 2 genetically identical diploid ($2n$) daughter cells; metaphase is the best stage to study chromosome morphology and count; anaphase is the stage where sister chromatids separate.
33. **Colchicine** (alkaloid from *Colchicum autumnale*) inhibits mitotic spindle microtubule formation, arresting cell division at **metaphase**; widely used to induce artificial **polyploidy** in plant breeding.
34. **Meiosis** (reductional division) occurs in germ cells, producing 4 genetically diverse haploid (n) daughter cells; chromosome number is halved during **Anaphase-I**.
35. **Prophase-I of Meiosis has 5 stages**: Leptotene (bouquet stage) → **Zygotene** (synapsis, synaptonemal complex formation) → **Pachytene** (**crossing over** between non-sister chromatids via enzyme **recombinase**) → **Diplotene** (**chiasmata** formation) → Diakinesis (terminalisation of chiasmata).

Chapter 03: Animal Biology and Husbandry

🎯 Consolidated Must-Score Facts

1. **Porifera** (sponges) exhibit cellular level of organisation, have thousands of minute incurrent pores (**Ostia**) and a single large excurrent opening (**Osculum**), and maintain a unique water canal system lined by flagellated **Choanocytes / Collar cells**.
2. **Cnidaria (Coelenterata)** possess stinging capsules called **Cnidoblasts / Nematocysts** for defence and prey capture; show radial symmetry and tissue-level organisation; exhibit alternation of generation (**Metagenesis**) between sessile cylindrical **Polyp** and free-swimming umbrella-like **Medusa** (e.g. *Hydra*, *Aurelia* jellyfish, *Physalia* Portuguese man-of-war). Corals (*Anthozoa*) secrete massive $CaCO_3$ skeletons of organic origin. *Hydra* has **no blood but respires** via simple diffusion.
3. **Platyhelminthes** (flatworms) are triploblastic, acoelomate, dorso-ventrally flattened worms; excrete and osmoregulate via specialized **Flame cells (Protonephridia)** (e.g. *Planaria*, *Taenia solium* pork tapeworm, *Fasciola hepatica* liver fluke). Tapeworms are **hermaphrodites** (monoecious).
4. **Aschelminthes / Nematoda** (roundworms) possess a **Pseudocoelom** (false body cavity); sexually dimorphic (separate sexes); examples: *Ascaris lumbricoides* (intestinal roundworm), *Wuchereria bancrofti* (filarial worm causing elephantiasis transmitted by *Culex* mosquito), *Dracunculus medinensis* (guinea worm, a nematode, NOT an annelid!).
5. **Annelida** (segmented worms) exhibit true **metameric segmentation**, a closed circulatory system, and excrete via coiled tubular **Nephridia**; earthworms breathe cutaneously across moist glandular skin; possess **4 to 5 pairs of lateral hearts** (aortic arches in segments 7–11 or 7, 9, 12, 13).
6. **Arthropoda** is the **largest phylum** of the animal kingdom (over two-thirds of all named species, >1 million insects); characterized by **chitinous exoskeleton**, jointed appendages, open circulatory system with a haemocoel, and excretion via **Malpighian tubules** (insects), **Green glands / Antennal glands** (crustaceans/prawns), or **Coxal glands** (spiders/scorpions).
7. **Insects (Class Insecta / Hexapoda)**: 3 main body segments (**Head, Thorax, Abdomen**), **3 pairs of legs (6 legs)** attached to the thorax, usually 2 pairs of wings, 1 pair of antennae, tracheal respiration via spiracles. Examples: Cockroach, Honeybee, Mosquito, Butterfly, Cicadas, Froghoppers (spittlebugs), Pond skaters (*Gerridae*), Termites (white ants, Isoptera).
8. **Arachnids (Class Arachnida)**: 2 body regions (**Cephalothorax and Abdomen**), **4 pairs of walking legs (8 legs)**, **zero wings and zero antennae**; mouth appendages are **chelicerae** (fangs) and **pedipalps**. Examples: Spiders, Scorpions, Ticks, Mites. Ticks and mites have unsegmented fused bodies.
9. **Spider Silk (Gossamer Silk)**: Protein fibre spun by spiders, renowned for being both strong and elastic (often described as stronger than steel). Spiders practice **sexual cannibalism** (female kills and eats the male before, during, or after copulation).
10. **Peacock Tarantula (*Poecilotheria metallica*)**: Critically Endangered, arboreal spider endemic to deciduous forests of Andhra Pradesh / Pakkam Malai; it is an **insectivorous arachnid, NOT a crustacean**.
11. **Mosquito Mouthparts & Posture**: Only **female mosquitoes suck blood** (mouthparts modified to pierce and suck: labium sheath, labrum feeding tube, paired mandibles and maxillae forming piercing stylet); **males have mouthparts adapted to suck flower/plant juices**. Females have larger wings. **Female *Anopheles rests at an angle (~45°)* to the surface with abdomen pointing up;*** Culex *rests parallel** to the surface.
12. **Bioluminescence in Glowworms**: Larvae and adults of *Arachnocampa luminosa* / fireflies emit blue-green light from abdominal light organs via the enzyme **Luciferase** acting on luciferin substrate.
13. **Beneficial vs Harmful Insects**: *Bombyx mori* (silk), *Laccifer / Kerria lacca* (lac resin), *Apis mellifera* (honey/wax) are beneficial. ***Helicoverpa armigera*** (cotton bollworm/corn earworm) is a polyphagous destructive agricultural pest.
14. **Parasitoid species**: Larvae develop inside/on a host and ultimately kill it (e.g. parasitic wasps, parasitic flies, certain carabid beetles). Centipedes are generalist predators; termites are detritivores.
15. **Detritivores & Filter Feeders**:
16. **Ant Biology**: Ants inject **Formic acid** ($HCOOH$, simple carboxylic acid) when they bite. Ants have the **largest brain relative to body size** in the animal kingdom. Leafcutter ants (*Attini*) actively **cultivate fungi** for food.
17. **Mollusca** is the second largest animal phylum; soft-bodied, unsegmented animals usually enclosed in a calcareous shell ($CaCO_3$); possess a sensory rasping tongue-like feeding organ called **Radula** and a fleshy muscular foot; true pearls are harvested from the marine pearl oyster (***Pinctada vulgaris***, class Bivalvia/Pelecypoda).
18. **Cephalopods**: **Octopus (Devilfish)** has 8 arms with suckers, camera eyes, and no external shell. **Colossal Squid (*Mesonychoteuthis hamiltoni*)** and **Giant Squid (*Architeuthis dux*)** are the largest living invertebrates.
19. **False Fish vs True Fish (Must-Score Fact)**:

20. **Fishes in Frozen Lakes**: Lakes freeze only at the surface; because water reaches maximum density at 4°C, the bottom water remains liquid ($>0^\circ\text{C}$), allowing fish and aquatic life to survive under the ice shield. Fish die outside water because gills collapse and cannot absorb gaseous oxygen.
21. **Echinodermata**: Exclusively marine spiny-skinned animals; exhibit secondary radial symmetry (bilateral in free-swimming larvae, pentamerous radial in adults); possess an ambulacral **Water Vascular System** operating suction-cup **Tube feet (Podia)**; **all echinoderms are NOT viviparous — they are typically oviparous with external fertilisation**.
22. **Animal heart chambers**: **Fish** = 2 chambers (1 atrium + 1 ventricle; single venous circulation); **Amphibians & most Reptiles** = 3 chambers (2 atria + 1 ventricle; mixed blood tolerated); **Crocodiles, Birds & Mammals** = **4 chambers** (2 atria + 2 ventricles; complete double circulation).
23. **Amphibian Metamorphosis**: Tadpoles are aquatic and herbivorous; metamorphosis into carnivorous frogs involves **dramatic shortening of the intestine** (controlled by thyroid hormone) to process meat diets. Frogs are cold-blooded and breathe via skin and lungs.
24. **Reptiles & Snakes**:
25. **Poisonous Species Across Taxa**: Poisonous species occur among **butterflies** (African Giant Swallowtail, Monarch — sequester plant alkaloids), **fishes** (Puffer fish Tetraodontidae with tetrodotoxin; Reef stonefish most venomous fish), and **frogs** (poison dart frogs with skin alkaloid secretions).
26. **Herbivorous Exceptions**: Adult **Green sea turtles** (*Chelonia mydas*) are strictly herbivorous; **Parrot fish** graze on algae; **Dugong / Sea cow** is a large herbivorous marine mammal. (Dogs are carnivores/omnivores, NOT herbivores).
27. **Flightless Birds & Aves**: **Kiwi** (*Apteryx*) is a small flightless bird endemic to New Zealand. **White Leghorn** is an exotic Mediterranean breed of fowl (chicken).
28. **Nocturnal Animals**: Highly developed hearing, smell, and adapted eyesight: Mosquitoes, Bats, Owls, Kiwis, Aye-aye (largest nocturnal primate using echolocation), **Malabar civet** (*Viverra civettina*). (Lion-tailed macaque is strictly diurnal; Sambar deer is crepuscular/nocturnal).
29. **Poikilothermic (Cold-blooded)**: Fish, Amphibians, Reptiles (body temperature fluctuates with environment).
30. **Homeothermic (Warm-blooded)**: Birds and Mammals (maintain stable internal body temperature via homeostasis).
31. **Respiratory organs across animal taxa**: Cutaneous (Earthworms, Frogs), Branchial (Fish, Tadpoles), Tracheal (Insects), Book lungs (Spiders, Scorpions), Book gills (Horseshoe crab *Limulus*), Pulmonary (Adult frogs, Reptiles, Birds, Mammals via blowholes in Cetaceans).
32. **Kangaroo rat** (*Dipodomys*) can survive its entire life without drinking water; produces the **most concentrated, almost solid urine** among all mammals and relies on metabolic water.
33. **Camels**: Store water in their **intestine** (can drink $>100\text{ litres}$ at once) and store energy as **fat in the hump**; have broad padded 2-toed feet; lack an Omasum (pseudoruminant).
34. **Gallbladder Absence**: Animals that do **not possess a gallbladder** include: all members of the deer family (except musk deer), equines (horses, zebras), camels, giraffes, elephants, rhinoceroses, whales, rats, and doves.
35. **Hooves vs Claws**: Horses, zebras, deer, cattle have hooves; **Jackal (*Canis aureus*)** belongs to Canidae and has **padded feet with nails/claws, NO hooves**.

Chapter 04: Nutrition, Vitamins, Diseases and Medicine

🎯 Consolidated Must-Score Facts

1. **Calcium** is the most abundant mineral element in the human body (~1.5% of body weight; 99% in bones and teeth as hydroxyapatite crystals); **Iron** is the most abundant essential micromineral (3–4 grams, primarily inside haemoglobin).
2. **Glucose** (dextrose / grape sugar) and **Fructose** (fruit sugar / levulose) are functional isomers with identical empirical formula ($C_6H_{12}O_6$); **Fructose is the sweetest naturally occurring sugar**.
3. **Sucrose** (table / cane sugar) is a non-reducing disaccharide composed of **Glucose + Fructose**; **Lactose** (milk sugar) is **Glucose + Galactose**; **Maltose** (malt sugar) is **Glucose + Glucose**.
4. **Starch** (plant storage carbohydrate) and **Glycogen** ("animal starch", stored in liver and skeletal muscles) are polymers of α-D-glucose digested by human amylases; **Cellulose** is a structural polymer of β-D-glucose that humans **cannot digest** due to the total absence of the enzyme cellulase (acts as roughage / dietary fibre).
5. **Kwashiorkor** is caused by severe **protein deficiency with adequate calorie intake**; characterized by pitting **oedema**, swollen "pot belly", and "flaky-paint" dermatosis; **Marasmus** is caused by combined **protein-energy malnutrition (PEM)** in infants $<1\text{ year}$; characterized by extreme emaciation, dry wrinkled skin ("little old man" face), and **absence of oedema**.
6. **Fat-Soluble Vitamins**: **A, D, E, K**; dissolve in organic solvents and fats, stored in the liver and adipose tissues; excessive intake causes toxic **Hypervitaminosis**.
7. **Water-Soluble Vitamins**: **B-Complex group and Vitamin C**; not stored in large amounts in the human body (readily excreted in urine, require daily dietary intake); **Vitamin B12 is the unique exception** (stored in the human liver for 3–5 years).
8. **Vitamin A (Retinol)**: Essential for rhodopsin synthesis in retinal rods; deficiency causes **Nyctalopia (Night Blindness)**, **Xerophthalmia** (dry, keratinized cornea), and **Bitot's spots**; rich sources: Cod-liver oil, liver, egg yolk, and plant precursors like **β-Carotene** in carrots, mangoes, and papayas.
9. **Vitamin D (Calciferol / Sunshine Vitamin)**: Synthesised in human skin upon exposure to solar UV-B radiation from 7-dehydrocholesterol; acts as a pro-hormone promoting intestinal absorption of calcium and phosphorus; deficiency causes **Rickets** in children (bowed legs, pigeon chest) and **Osteomalacia** in adults (soft, fragile bones).
10. **Vitamin E (Tocopherol / Anti-Sterility Vitamin)**: Powerful lipid-soluble biological **antioxidant** protecting cell membranes from free radical oxidation; richest natural dietary source is **Wheat-germ oil**, almonds, and sunflower oil.
11. **Vitamin K (Phylloquinone / Naphthoquinone)**: Required in the liver as a cofactor for the post-translational carboxylation of clotting factors **II (Prothrombin), VII, IX, and X**; deficiency causes prolonged bleeding time and haemorrhage; synthesized by symbiotic colon bacteria (*E. coli*); rich plant source: **Alfalfa** and dark green leafy vegetables.
12. **Vitamin C (Ascorbic Acid / Anti-Scurvy Vitamin)**: Required for the enzymatic hydroxylation of proline and lysine during **Collagen synthesis**; deficiency causes **Scurvy** (bleeding and spongy gums, loose teeth, poor wound healing, petechial haemorrhages); destroyed by prolonged boiling and cooking heat; richest sources: **Amla (Indian gooseberry)**, guava, citrus fruits (lemon, orange); **absent in milk, meat, and eggs**.
13. **Vitamin B1 (Thiamine)**: Active coenzyme TPP (Thiamine Pyrophosphate) required in carbohydrate metabolism; deficiency causes **Beriberi** (Dry Beriberi = peripheral neuropathy; Wet Beriberi = congestive heart failure and oedema); **eating polished / milled parboiled rice strips the thiamine-rich outer aleurone layer**, precipitating beriberi.
14. **Vitamin B2 (Riboflavin)**: Precursor of coenzymes FMN and FAD; imparts characteristic yellow colour to cow milk and whey; deficiency causes **Ariboflavinosis**: **Cheilosis** (painful cracks at the corners of the mouth), **Glossitis** (magenta-red inflamed tongue), and keratitis.
15. **Vitamin B3 (Niacin / Nicotinic Acid)**: Precursor of NAD and NADP; synthesised in small quantities from the essential amino acid **Tryptophan**; deficiency causes **Pellagra**, classically remembered by the **"4 Ds"**: **Dermatitis** (Casal's necklace pattern on sun-exposed skin), **Diarrhoea**, **Dementia**, and **Death**; prevalent in populations relying exclusively on maize / corn diets (maize is deficient in tryptophan and contains bound niacytin).
16. **Vitamin B5 (Pantothenic Acid)**: Essential constituent of **Coenzyme-A (CoA)**; deficiency causes "Burning Feet Syndrome" (*Gopalan's syndrome*).
17. **Vitamin B6 (Pyridoxine)**: Coenzyme for amino acid transamination and decarboxylation; required for synthesis of haemoglobin, serotonin, and GABA; deficiency causes peripheral neuropathy, microcytic anaemia, and convulsions in infants; *Lucent note*: **Vitamin B6 helps in remembering dreams**.

18. **Vitamin B7 (Biotin / Vitamin H)**: Sulfur-containing coenzyme for carboxylation reactions; raw egg white contains the protein **Avidin**, which binds biotin tightly and blocks absorption, causing induced biotin deficiency.
19. **Vitamin B9 (Folic Acid / Folate)**: Required for DNA synthesis and purine/thymidylate production; deficiency causes **Megaloblastic Anaemia**; maternal folate deficiency in periconceptional period causes fatal **Neural Tube Defects (Spina Bifida and Anencephaly)** in newborns.
20. **Vitamin B12 (Cyanocobalamin)**: The only vitamin containing a metallic trace element (**Cobalt**); absent in ordinary plant foods; synthesized exclusively by microorganisms; absorbs in the terminal ileum mediated by **Castle's Intrinsic Factor** secreted by gastric parietal cells; deficiency causes **Pernicious Anaemia** (megaloblastic anaemia + irreversible subacute combined degeneration of the spinal cord).
21. **Bacterial Diseases**:
22. **Protozoan Diseases**:
23. **Cancer Biology & Classification**:
24. **Diabetes Mellitus vs Diabetes Insipidus**:
25. **Vaccine Classification & Modern Immunology**:
26. **Clinical Diagnostic Tests & Instruments Drill**:
27. **Chemical Elemental Composition of the Human Body**:
28. **Oxygen ($\sim 65\%$)**: Most abundant element by mass due to the vast water content ($\sim 60–65\%$ of body weight).
29. **Carbon ($\sim 18.5\%$)**: Backbone of all organic macromolecules (carbohydrates, lipids, proteins, nucleic acids).
30. **Hydrogen ($\sim 9.5\%$)**: Most abundant element by **total atom count** (each water molecule H_2O contains two hydrogen atoms).
31. **Nitrogen ($\sim 3.2\%$)**: Integral constituent of proteins, amino acids, DNA, RNA, ATP, and creatine.
32. **Calcium ($\sim 1.5\%$)**: Most abundant mineral; structural hydroxyapatite in bone and teeth, blood coagulation, muscle contraction.
33. **Phosphorus ($\sim 1.0\%$)**: Calcium phosphate in bones, phospholipid membranes, phosphodiester bonds of DNA/RNA, and high-energy ATP.
34. **Macronutrients: Carbohydrates, Proteins & Fats**:
35. **Carbohydrates**: Primary and immediate source of physiological energy; stored as **Glycogen** in liver (~100 g) and skeletal muscles (~400 g). Excess carbohydrates are converted to neutral fats via **lipogenesis** and deposited in subcutaneous adipose tissue.

Chapter 05: Digestion, Respiration and Excretion

🎯 Consolidated Must-Score Facts

1. **Tooth Enamel** is the **hardest substance in the human body**; secreted by specialized epithelial cells called **Ameloblasts**; composed of 96% inorganic mineral crystals of **Calcium Hydroxyapatite**; dentine is secreted by **Odontoblasts**.
2. **Trypsin** and **Chymotrypsin** are proteolytic enzymes produced by the **pancreas** and active in the alkaline medium of the **small intestine**; they are **never secreted by the stomach**.
3. **Bile contains NO digestive enzymes whatsoever**; contains bile salts (**Sodium Glycocholate and Sodium Taurocholate**) which **emulsify fats** (breaking large lipid globules into micro-droplets) and bile pigments (**Bilirubin and Biliverdin**).
4. **Tidal Volume (TV)**: Volume of air inspired or expired during a normal quiet breath = **$\mathbf{\sim 500\text{ mL}}$** in healthy adults.
5. **Inspiratory Reserve Volume (IRV)**: Extra volume inspired forcefully after normal tidal inspiration = $2500–3000\text{ mL}$.
6. **Expiratory Reserve Volume (ERV)**: Extra volume expired forcefully after normal tidal expiration = $1000–1100\text{ mL}$.
7. **Residual Volume (RV)**: Volume of air remaining permanently inside lungs even after maximal forced expiration = **$\mathbf{1100–1200\text{ mL}}$**; prevents alveolar collapse; **cannot be measured by a spirometer**.
8. **Vital Capacity (VC)**: Maximum volume of air exhaled after maximal forced inspiration: $VC = TV + IRV + ERV \approx \mathbf{3500–4500\text{ mL}}$.
9. **Oxygen Transport in Blood**: **$\mathbf{97\%}$ is transported bound to Haemoglobin as Oxyhaemoglobin** inside red blood cells; only **$\mathbf{3\%}$** is dissolved in blood plasma.
10. **Carbon Dioxide Transport in Blood**:
11. **$\mathbf{\sim 70\%}$ is transported as Bicarbonate ions ($\mathbf{HCO_3^-}$)** dissolved in blood plasma.
12. **$\mathbf{\sim 20–25\%}$ is transported bound to haemoglobin as Carbamino-haemoglobin**.
13. **$\mathbf{\sim 7\%}$ is transported dissolved in blood plasma**.
14. **Bohr Effect**: Lowering of blood pH (acidosis) or elevated carbon dioxide tension (pCO_2) shifts the oxyhaemoglobin dissociation curve to the **right**, decreasing haemoglobin's oxygen affinity and promoting rapid unloading of oxygen to respiring tissues.
15. **Carbon Monoxide (CO) Poisoning**: Carbon monoxide has an affinity for haemoglobin **$\mathbf{200–250\text{ times greater}}$ than oxygen**; forms extremely stable **Carboxyhaemoglobin ($COHb$)**, permanently blocking oxygen binding sites, causing severe cellular asphyxiation, tissue hypoxia, and cherry-red skin discoloration.
16. **Regulation of Respiration**: Primary respiratory rhythm centre is located in the **Medulla Oblongata** of the brain stem; **Pneumotaxic centre** is in the **Pons Varolii**; central chemoreceptors are exquisitely sensitive to arterial **excess of $\mathbf{CO_2}$ and Hydrogen ions ($\mathbf{H^+}$)**, **not to low oxygen**.
17. **Ammonotelic**: Excrete highly toxic **Ammonia** requiring copious water ($300–500\text{ mL/g}$); e.g. bony fish, aquatic amphibians (tadpoles), aquatic invertebrates.
18. **Ureotelic**: Excrete less toxic **Urea**; synthesised in the liver via the **Ornithine Cycle / Krebs-Henseleit Cycle** from ammonia and CO_2; e.g. mammals, adult amphibians, marine sharks.
19. **Uricotelic**: Excrete non-toxic, insoluble **Uric Acid** as a semi-solid paste/pellet with minimal water loss; e.g. birds, terrestrial reptiles, insects, land snails.
20. **Malpighian Body (Renal Corpuscle)**: Composed of a tuft of capillaries called the **Glomerulus** encased inside a double-walled cup called **Bowman's Capsule**.
21. **Ultrafiltration**: Glomerular capillary blood pressure forces water, glucose, amino acids, ions, and urea across the trilaminar filtration membrane into Bowman's capsule, forming **Glomerular Filtrate (180 litres/day; GFR = 125 mL/min)**; blood cells and plasma proteins (albumin) are retained.
22. **Selective Reabsorption in the Proximal Convoluted Tubule (PCT)**: PCT is lined by simple cuboidal brush-border epithelium with microvilli; **reabsorbs $\mathbf{70–80\%}$ of all filtered water and electrolytes, and $\mathbf{100\%}$ of filtered glucose and amino acids** via active transport.
23. **Loop of Henle & Counter-Current Multiplier**: Hairpin loop maintaining high osmotic medullary gradient; descending limb is permeable to water and impermeable to salts; ascending limb is impermeable to water and actively pumps out NaCl; operates

alongside **Vasa Recta** to produce concentrated hypertonic urine.

24. **Hormonal Regulation of Excretion**:
25. **Antidiuretic Hormone (ADH / Vasopressin)**: Secreted by posterior pituitary; acts on DCT and collecting ducts to insert aquaporins, promoting water reabsorption; deficiency causes **Diabetes Insipidus** (profuse dilute urine).
26. **Renin-Angiotensin-Aldosterone System (RAAS)**: Juxtaglomerular apparatus (JGA) releases **Renin** in response to low blood pressure; renin converts angiotensinogen $\rightarrow$ angiotensin-I $\rightarrow$ angiotensin-II (powerful vasoconstrictor); stimulates adrenal cortex to release **Aldosterone**, increasing Na^+ and water reabsorption.
27. **Atrial Natriuretic Factor (ANF)**: Secreted by heart atria in response to high blood pressure; promotes sodium excretion (natriuresis) and vasodilation, antagonising RAAS.
28. **Renal Calculi (Kidney Stones)**: Solid crystalline deposits formed inside renal pelvis and calyces, primarily composed of insoluble **Calcium Oxalate (CaC_2O_4)** crystals.
29. **Gout**: Painful metabolic arthritis caused by the deposition of sharp **Monosodium Urate / Uric Acid crystals** in synovial joints (most commonly the metatarsophalangeal joint of the big toe) due to hyperuricaemia.
30. **The Human Mouth & Mechanical Mastication**:
31. **Human Dentition Characteristics**:
32. **Thecodont**: Every tooth is embedded in a deep bony jaw socket (alveolus).
33. **Diphyodont**: Two sets of teeth during lifetime: deciduous / milk teeth (20 teeth) replaced by permanent adult teeth (32 teeth).
34. **Heterodont**: Four morphologically and functionally distinct types of teeth:
35. **Incisors (I)**: Chisel-shaped, for biting and cutting food.

Chapter 06: Blood, Heart, Circulation and Lymph

🎯 Consolidated Must-Score Facts

1. **Erythrocytes (Red Blood Cells / RBCs)**: Normal count is **5.0–5.5 million / $\mathbf{mm^3}$** in males and 4.5–5.0 million / mm^3 in females; mature mammalian RBCs are **biconcave, circular, and enucleated** (lack nucleus, mitochondria, and Golgi bodies; **Camel and Llama are exceptions: their RBCs are oval and nucleated**).
2. **Haemoglobin (Hb)**: Conjugated chromoprotein; normal concentration is **$14–17\text{ g/dL}$** in adult men and **$12–15\text{ g/dL}$** in women; consists of the protein globin and 4 iron-porphyrin **Haem** groups containing **ferrous iron ($\mathbf{Fe^{2+}}$)**; imparts the red colour to blood.
3. **Methemoglobin & Blue Baby Syndrome**: When ferrous iron (Fe^{2+}) of haemoglobin is oxidized into ferric iron (Fe^{3+}), it forms **Methemoglobin**, which cannot bind or carry oxygen; infants ingesting groundwater contaminated with **Nitrates (NO_3^-)** suffer from **Methemoglobinemia (Blue Baby Syndrome)**, presenting with acute cyanosis and chocolate-brown blood.
4. **Leukocytes (White Blood Cells / WBCs)**: Total count is **$6,000–8,000/\mathbf{mm^3}$**; nucleated, lack haemoglobin; divided into **Granulocytes** and **Agranulocytes**; the normal ratio of RBC to WBC is **$600:1$**.
5. **Neutrophils (60–65%)**: Most numerous; multi-lobed nucleus; primary phagocytic first-responders that engulf bacteria and cell debris.
6. **Eosinophils / Acidophils (2–3%)**: Bilobed nucleus; release histaminase; counts surge in **allergic reactions and intestinal helminth parasitic infections**.
7. **Basophils (0.5–1%)**: Least numerous; secrete **Histamine** (vasodilator in inflammation), **Serotonin**, and **Heparin** (natural anticoagulant).
8. **Monocytes (6–8%)**: Largest WBCs; bean/kidney-shaped nucleus; enter tissues to transform into massive phagocytic **Macrophages**.
9. **Lymphocytes (20–25%)**: Large rounded nucleus; responsible for specific adaptive immunity: **B-lymphocytes** (produce humoral antibodies) and **T-lymphocytes** (mature in the thymus, execute cell-mediated immunity).
10. **Thrombocytes (Blood Platelets)**: Normal count is **$\mathbf{1.5–4.5\text{ lakh}}/\mathbf{mm^3}$**; non-nucleated protoplasmic cell fragments budded from giant marrow **Megakaryocytes**; lifespan 7–10 days; aggregate at wound sites and release **Thromboplastin** to trigger blood clotting; depleted drastically in dengue fever.
11. **Blood Clotting (Coagulation Cascade)**:
12. **Vitamin K** is indispensable for hepatic synthesis of four essential clotting factors: **Factor II (Prothrombin), Factor VII, Factor IX, and Factor X**.
13. **Heparin** is a natural conjugated polysaccharide anticoagulant secreted by basophils and mast cells; prevents intravascular clotting inside healthy blood vessels.
14. **ABO Blood Group System**: Discovered by Austrian biologist **Karl Landsteiner (1900–1901)**; blood groups (A, B, AB, O) are determined by specific oligosaccharide **antigens on the surface of RBC membranes** and complementary **antibodies in blood plasma**:
15. **Group A**: Has A antigen on RBCs; anti-B antibody in plasma.
16. **Group B**: Has B antigen on RBCs; anti-A antibody in plasma.
17. **Group AB**: Has **both A and B antigens** on RBCs; **NEITHER anti-A nor anti-B antibody** in plasma.
18. **Group O**: Has **NEITHER A nor B antigen** on RBCs; **BOTH anti-A and anti-B antibodies** in plasma.
19. **Rh Factor (Rhesus Antigen / Antigen D)**: Discovered by **Karl Landsteiner and Alexander S. Wiener (1940)** in the Rhesus macaque monkey; present on RBCs of ~85% of humans (Rh-positive); ~15% are Rh-negative.
20. **Erythroblastosis Fetalis (Haemolytic Disease of the Newborn)**: Occurs when an **Rh-negative mother** conceives an **Rh-positive fetus**; fetal Rh antigens leak into maternal blood during first delivery, sensitizing the mother to produce maternal anti-Rh antibodies (**IgG class**); during a subsequent Rh-positive pregnancy, maternal anti-Rh IgG crosses the placenta and hemolyses fetal erythrocytes, causing severe fatal anaemia, kernicterus, and hydrops fetalis; prevented by administering **Anti-D Rh-immunoglobulin (RhoGAM)** to the mother within 72 hours of delivering an Rh-positive baby.
21. **Bombay Blood Group (hh Phenotype)**: Rare blood phenotype discovered in Mumbai by **Dr. Y. M. Bhende (1952)**; individuals lack the universal precursor **H-gene**, so cannot biosynthesise A or B antigens; tests as Group O in routine ABO grouping, but

serum contains anti-A, anti-B, and **anti-H antibodies**; can donate blood to anyone in the ABO system, but can **receive blood ONLY from another Bombay blood group donor**.

22. **Tricuspid Valve**: Possesses 3 muscular cusps; guards the right atrio-ventricular orifice between the right atrium and right ventricle.
23. **Bicuspid Valve (Mitral Valve)**: Possesses 2 muscular cusps; guards the left atrio-ventricular orifice between the left atrium and left ventricle.
24. **Semilunar Valves**: Guard the exit of the right ventricle into the **Pulmonary Artery** (Pulmonary semilunar valve) and the exit of the left ventricle into the **Aorta** (Aortic semilunar valve); prevent backflow into ventricles during diastole.
25. **Nodal Conduction System (Myogenic Autonomic Rhythmicity)**:
26. **SA Node (Sino-Atrial Node / "Heart of Hearts")**: Located in the upper-right corner of the right atrium near the entry of the superior vena cava; generates intrinsic rhythmic action potentials at **70–75 beats/minute**; functions as the natural **Pacemaker of the Heart**.
27. **AV Node (Atrio-Ventricular Node)**: Located in the lower-left corner of the right atrium near the inter-atrial septum; acts as a relay station, introducing a crucial **0.1-second delay** to allow atria to empty completely before ventricular contraction begins; called the **Pacesetter**.
28. **Bundle of His (AV Bundle)**: Specialized muscle fascicle descending down the interventricular septum, bifurcating into Right and Left Bundle Branches.
29. **Purkinje Fibres**: Microscopic conduction fibres spreading rapidly throughout the ventricular myocardium to trigger coordinated contraction from the cardiac apex upward.
30. **Atrial Systole**: 0.1 second (atria contract, topping up ventricles).
31. **Ventricular Systole**: 0.3 second (ventricles contract at high pressure, pumping blood into aorta and pulmonary artery).
32. **Joint Diastole (Complete Cardiac Diastole)**: 0.4 second (all four chambers relax simultaneously and refill passively).
33. **Stroke Volume**: Volume of blood pumped out by each ventricle during each individual cardiac cycle = **$\mathbf{\sim 70\text{ mL}}$**.
34. **Cardiac Output**: Volume of blood pumped by each ventricle per minute:
35. **Heart Sounds Auscultated via Stethoscope**:

Chapter 07: Nervous System, Hormones, Reproduction and Support

🎯 Consolidated Must-Score Facts

1. **Cerebrum**: Largest part of the human brain (~80% of brain mass); seat of intelligence, conscious thought, memory, reasoning, voluntary motor control, speech (Broca's area), and sensation; right and left cerebral hemispheres are joined internally by a broad transverse curved band of white nerve fibres called the **Corpus Callosum**.
2. **Hypothalamus**: Situated at the base of the diencephalon; acts as the body's **Thermostat** (regulates core body temperature); controls hunger, thirst, satiety, sleep-wake cycles, emotional responses (anger, fear, pleasure via the limbic system), and regulates the pituitary gland via releasing and inhibiting neurohormones.
3. **Cerebellum**: Second largest part of the brain; displays an internal tree-like branching pattern of white matter called **Arbor Vitae** ("Tree of Life"); coordinates voluntary muscular contractions, maintains posture, and controls **physical equilibrium and body balance**; alcohol intoxication temporarily depresses the cerebellum, causing muscular uncoordination and a staggering gait.
4. **Medulla Oblongata**: Continues inferiorly into the spinal cord; houses vital autonomous reflex centres controlling **respiratory rhythm, cardiac rate, arterial blood pressure (vasomotor centre)**, as well as protective reflexes: swallowing, coughing, sneezing, vomiting, and salivation.
5. **Endocrine Glands (Ductless Glands)**: Secretions are poured directly into circulating blood; target distant organs; e.g. Pituitary, Thyroid, Parathyroid, Adrenal, Pineal, Thymus.
6. **Exocrine Glands**: Possess tubular ducts that deliver secretions locally onto an epithelial surface; e.g. Salivary, Sweat, Sebaceous, Lacrimal (tear), Mammary glands.
7. **Heterocrine / Mixed Glands**: The **Pancreas** acts as both an exocrine gland (acinar cells secrete digestive enzymes via the pancreatic duct) and an endocrine gland (Islets of Langerhans secrete insulin and glucagon into blood).
8. **The Pituitary Gland (Hypophysis / Master Gland)**:
9. **The Thyroid Gland**:
10. **The Parathyroid Glands**:
11. **The Adrenal Glands (Suprarenal Glands / 4S & 3F Glands)**:
12. **The Pineal Gland (Epiphysis Cerebri)**: Located on the dorsal roof of the diencephalon; secretes **Melatonin**, which regulates the 24-hour diurnal **Circadian Rhythm (Sleep-Wake Cycle)**, core body temperature, and delays premature puberty.
13. **The Thymus Gland**: Located behind the sternum in the anterior mediastinum; secretes **Thymosins**, which stimulate the maturation and differentiation of **T-lymphocytes**; acts as the primary "training school" of cellular immunity; prominent in infants, undergoes progressive age-related atrophy after puberty, leading to weakened immunity in the elderly.
14. **Male Reproductive System**:
15. **Female Reproductive System & The Menstrual Cycle**:
16. **Implantation, Placenta & hCG**:
17. **Contraception & Reproductive Health**:
18. **The Human Skeletal System (206 Bones in Adult)**:
19. **Joints of the Human Skeleton**:
20. **Connective Support Tissues: Tendon vs Ligament**:
21. **Muscular Physiology & Contraction**:
22. **The Neuron (Structural & Functional Unit)**:
23. **Phylogenetic Distribution Fact**: All multicellular animals possess neurons **except Sponges (Porifera)** and Placozoans. Plants completely lack neurons.
24. **Peripheral Nerve Geography**:
25. **Cranial Nerves**: **12 pairs** originating directly from the ventral base of the brain and brainstem, innervating structures of the head, neck, and viscera (e.g. Vagus nerve X).
26. **Spinal Nerves**: **31 pairs** of mixed nerves emerging bilaterally from the spinal cord segments:
27. **8 Cervical pairs** ($C_1–C_8$)

28. **12 Thoracic pairs** ($T_1–T_{12}$)
29. **5 Lumbar pairs** ($L_1–L_5$)
30. **5 Sacral pairs** ($S_1–S_5$)
31. **1 Coccygeal pair** (Co_1)
32. **Autonomic Nervous System (ANS) Dual Antagonism**:
33. **Sympathetic Division ("Fight or Flight")**: Activated during stress, emergency, and intense exercise; increases heart rate and cardiac contractility, dilates bronchioles, dilates pupils, inhibits gastrointestinal peristalsis and salivation, diverts blood flow from skin and viscera to skeletal muscles; utilizes **Norepinephrine / Noradrenaline** (adrenergic fibres).
34. **Parasympathetic Division ("Rest and Digest")**: Activated in relaxed, quiescent states; slows heart rate, constricts pupils, stimulates digestive secretions, peristalsis, and glycogen synthesis; mediated primarily via the **Vagus nerve (CN X)**; utilizes **Acetylcholine** (cholinergic fibres).
35. **Enteric Nervous System (ENS)**: Semi-autonomous intrinsic neural meshwork in the gastrointestinal submucosal and myenteric plexuses; termed the **"Second Brain"**.

Chapter 08: Plants, Agriculture and Plant Diseases

🎯 Consolidated Must-Score Facts

1. **Plant Taxonomy Hierarchy**: Categorized from higher to lower: **Kingdom $\rightarrow$ Phylum (Division) $\rightarrow$ Class $\rightarrow$ Order $\rightarrow$ Family $\rightarrow$ Genus $\rightarrow$ Species**. **Species** is the basic taxonomic unit consisting of individuals with fundamental morphological similarities capable of interbreeding.
2. **Augustin Pyramus de Candolle (A.P. de Candolle)**: Swiss botanist who **first coined the term "Taxonomy" (1813)** and first used internal **vascular tissue characteristics** to divide plants into *Vasculares* (with vascular bundles) and *Cellulares* (without vascular bundles). (Linnaeus is the "Father of Modern Taxonomy").
3. **Plant Tissues**: **Apical Meristem** (vertical height / primary growth); **Intercalary Meristem** (internode bases, re-growth of grazed grasses); **Lateral Meristem / Cambium** (secondary growth, increases **girth / diameter** of stems).
4. **Simple Permanent Tissues**: **Parenchyma** (living packing tissue, modified into **Aerenchyma** with large air chambers in aquatic hydrophytes for buoyancy); **Collenchyma** (living with pectin corner thickenings, provides mechanical flexibility); **Sclerenchyma** (dead, heavily lignified; commercial jute/hemp fibres and gritty sclereids/stone cells of guava and pear).
5. **Complex Vascular Tissues**:
6. **Ascent of Sap**: Driven by **Transpiration Pull** based on the **Cohesion-Tension Theory** of **Dixon and Joly (1894)**. Positive **Root Pressure** causes **Guttation** through marginal leaf **Hydathodes** in moist early mornings.
7. **Modified Underground Stems (Food Storage)**:
8. **Modified Storage Roots (NOT Stems)**:
9. **Aerial Stem Storage**: In **Sugarcane (*Saccharum officinarum*, Family Gramineae/Poaceae)**, sucrose is stored predominantly in the **aerial stalk / stem** (propagated by stem cuttings with nodes).
10. **The Pea Family (Fabaceae / Leguminosae)**: Dicotyledonous plants with root nodules harbouring nitrogen-fixing bacteria: **Pea (*Pisum sativum*, herb with leaf tendrils)**, **Groundnut / Peanut (*Arachis hypogaea*)**, **Horse-gram / Kulthi (*Macrotyloma uniflorum*)**, **Soybean (*Glycine max*)**.
11. **Spices & Floral Parts**:
12. **True Fruits vs False Fruits**:
13. **Fruit Classifications & Edible Anatomy**:
14. **Plant Pigments Master Fact**:
15. **Plant Odour & Pungency Chemistry**:
16. **Commercial Fibres & Economic Products**:
17. **Ecological Plant Adaptations**:
18. **Temperature Adaptations in Ecology**:
19. **Forensic Utility of Diatoms**: **Diatoms** are unicellular photosynthetic algae with indestructible silica shells (*frustules*); inhalation of water during drowning draws diatoms into the bloodstream and internal organs (bone marrow, liver); **presence of diatoms in closed organs is a definitive forensic test confirming death by drowning**.
20. **Insectivorous & Carnivorous Plants**: Grow in acidic, waterlogged bogs **severely deficient in nitrates/nitrogen**; photosynthesise carbohydrates normally, but trap and digest small insects using proteases to fulfill nitrogen demands:
21. **Photosynthesis Physiology**:
22. **Plant Respiration & Cold Storage**: Cold storage extends fruit and vegetable shelf life because **low temperatures slow down the rate of cellular respiration and enzymatic activity** (UPPCS 2021).
23. **Phytohormones Master List**:
24. **Biological Nitrogen Fixation (BNF) & Biofertilisers**:
25. **Crop Diseases & Nutritional Disorders**:
26. **Agronomic Management Facts for Nitrogen Use Efficiency (NUE)**:
27. **Split application** of nitrogenous fertilizers matching crop physiological demand stages.
28. **Non-Nitrogen-Fixing Crops Trap**: Paddy, wheat, sugarcane, amaranth, spinach, and purslane (kulfa) do **not** fix atmospheric nitrogen on their own and require microbial inoculants or chemical nitrogen supplementation!

29. **Nature of Photosynthesis**: Oxidation-reduction (redox) photochemical reaction in which water is oxidized to release molecular oxygen (O_2) and carbon dioxide is reduced to synthesize carbohydrates (glucose).
30. **Energy Conversion**: Solar radiation energy is transformed into **chemical potential energy** stored in carbohydrate molecular bonds.
31. **Global Photosynthetic Output**: Approximately **90% of Earth's total photosynthesis** is executed by marine and freshwater **algae and phytoplankton** (85% in marine oceans, 5% in freshwater rivers/lakes). Terrestrial forests and grasses contribute the remainder.
32. **Source of Evolved Oxygen**: The oxygen gas (O_2) liberated during photosynthesis **originates entirely from the photolysis of water (H_2O)**, proven via isotope tracer experiments (^{18}O), and **not from carbon dioxide**.
33. **Self-Replication**: Chloroplasts possess their own **extrachromosomal circular DNA and 70S RNA**, conferring autonomous self-replicating capacity (semiautonomous organelle).
34. **Chlorophyll Molecule**: Chelated porphyrin ring with a coordinated central **Magnesium (Mg)** ion. Absence of magnesium leads to severe interveinal chlorosis (yellowing). Elements in chlorophyll: C, H, O, N, Mg (**Calcium is NOT present!**).
35. **Absorption vs Action Spectrum**:

Chapter 09: Microbiology, Environment and Applied Science

🎯 Consolidated Must-Score Facts

1. **The Four Microbial Groups**:
2. **Bacterial Morphology & Gram Staining**:
3. **Endospores & Archaebacteria**:
4. **Viruses, Bacteriophages & Sub-Viral Agents**:
5. **Fungi: Yeasts, Moulds & Lichens**:
6. **Industrial Microbiology & Microbes in Human Welfare (NCERT Class 12 Master Pairs)**:
7. **Microbial Organic Acids & Industrial Enzymes**:
8. **Bioactive Molecules & Wonder Drugs (High-Yield UPPCS Match-Pairs)**:
9. **Antibiotics & Milestone Discoveries**:
10. **The Disease–Pathogen Pairs (Crucial Traps)**:
11. **Scientific Method & Great Medical Pioneers**:
12. **Sewage Treatment Plants (STP) & Water Quality Parameters**:
13. **Biogas (Gobar Gas) Technology**:
14. **Drinking Water Purification & Biological Contamination Markers**:
15. **Waste Management: Biodegradable vs Non-Biodegradable Pollutants**:
16. **Biomagnification vs Bioaccumulation**:
17. **Bioremediation & Microbial Environmental Clean-Up**:
18. **Biomedical Waste Color Coding (Ministry of Environment, Forest & Climate Change Rules)**:
19. **Major Research Institutes of Uttar Pradesh (Heavily Tested UPPCS Matches)**:
20. **National & International Apex Scientific Institutions**:
21. **Indian Scientific Legends & Inventions**:
22. **The Six Traditional Indian Seasons (*Shad Ritus*)**:
23. **Important Commemorative Days for Science, Health and Environment**:
24. **Discovery of Bacteria & Microscopy**:
25. **Autotrophic Bacteria & Photosynthetic Pigments**:
26. **Gram Stain & Peptidoglycan Cell Wall**:
27. **Mycoplasma (Smallest Living Cells)**:
28. **Synthetic Medium Culturing vs Viral Host Dependence**:
29. **Asexual Reproduction & Spore Formation**:
30. **Fungi Kingdom Characteristics**:
31. **Edible vs Toxic Mushrooms & Dimorphic Fungi**:
32. **Master Spectrum of Biological Interactions**:
33. **Mycorrhizal Association**:
34. **Lichens: Pioneer Organisms & Air Quality Indicators**:
35. **Ecosystem Trophic Architecture & Producers**:

Chapter 10: Physics Fundamentals, Measurement and Instruments

🎯 Consolidated Must-Score Facts

1. **BIPM 2019 SI Redefinition (Effective May 20, 2019)**: All 7 base units are now anchored to fixed numerical values of fundamental physical constants:
2. **New SI Prefixes (27th CGPM, Versailles, 2022)**:
3. **Solar Constant & TSIS-1 (NASA Consensus)**: Total Solar Irradiance (TSI) received at 1 AU outside Earth's atmosphere is standardized at **$1361.6 \pm 0.3\text{ W/m}^2 \approx 1.4\text{ kW/m}^2$** (measured by TSIS-1 / SORCE; varies by $\sim 0.1\%$ over the 11-year solar sunspot cycle).
4. **MEMS Accelerometers in Modern Tech (IAS 2023 Fact)**: Accelerometers sense dynamic acceleration and tilt. Crucial for: (1) Airbag deployment via instant deceleration detection, (2) Free-fall sensor (FFS) in laptops parking HDD heads to prevent drive crash, (3) Display auto-rotation between portrait and landscape in smartphones.
5. **Four Systems of Units**: CGS (cm, g, s), FPS ($foot, pound, second$ — British system; $1\text{ lb} = 453.59237\text{ g}$), MKS ($m, kg, s$), and SI (International System of Units, adopted at 11th CGPM Paris in 1960).
6. **Seven Fundamental Quantities**: Length (m), Mass (kg), Time (s), Electric Current (A), Thermodynamic Temperature (K), Luminous Intensity (cd), Amount of Substance (mol).
7. **Supplementary Units**: Plane angle (Radian, $rad = m/m$, dimensionless), Solid angle (Steradian, $sr = m^2/m^2$, dimensionless).
8. **22 SI Derived Units with Special Names**: Radian (rad), Steradian (sr), Hertz (Hz), Newton (N), Pascal (Pa), Joule (J), Watt (W), Coulomb (C), Volt (V), Farad (F), Ohm (Ω), Siemens (S), Weber (Wb), Tesla (T), Henry (H), Degree Celsius ($^\circ C$), Lumen (lm), Lux (lx), Becquerel (Bq), Gray (Gy), Sievert (Sv), Katal (kat).
9. **Katal (kat)**: SI unit of catalytic activity ($1\text{ kat} = 1\text{ mol/s}$ of substrate converted).
10. **Tesla (T)**: SI unit of magnetic flux density (B-field). $1\text{ T} = 1\text{ Wb/m}^2 = 1\text{ N}/(A\cdot m) = 1\text{ kg}\cdot s^{-2}\cdot A^{-1} = 10^4\text{ Gauss}$.
11. **Magnetic Field Strength (H) vs Flux Density (B)**: Unit of H is Ampere per metre (A/m); unit of B is Tesla (T).
12. **Electrical Resistivity (ρ)**: SI unit is **Ohm-metre** ($\Omega\cdot m$), from $R = \rho l / A \implies \rho = RA/l$.
13. **Electrical Conductance (G) & Conductivity (σ)**: Conductance unit is Siemens ($S = \Omega^{-1}$ or Mho); Conductivity unit is Siemens per metre (S/m or $\Omega^{-1}\cdot m^{-1}$).
14. **Capacitance**: SI unit is Farad ($F = C/V$). (Note: Faraday is a unit of electric charge: $1\text{ Faraday} \approx 96,485\text{ C}$).
15. **Radioactivity Units**: SI unit is **Becquerel** ($Bq = 1\text{ disintegration/s}$). Historical unit is **Curie** ($Ci = 3.7 \times 10^{10}\text{ Bq}$). Rutherford ($Rd = 10^6\text{ Bq}$).
16. **Radiation Absorbed Dose vs Dose Equivalent**: Absorbed dose is **Gray** ($Gy = 1\text{ J/kg} = 100\text{ rad}$). Equivalent/effective biological dose is **Sievert** ($Sv = 1\text{ J/kg} = 100\text{ rem}$).
17. **Light-Year (ly)**: Unit of **astronomical distance** (NOT time). Distance travelled by light in vacuum in 1 Julian year ($365.25\text{ days} \times 86,400\text{ s}$). $1\text{ ly} = 9.46073 \times 10^{15}\text{ m} \approx 9.461 \times 10^{15}\text{ m} \approx 9.46 \times 10^{12}\text{ km}$.
18. **Why Astronomical Distances are Measured in Light-Years**: The speed of light (c) is constant throughout the universe in a vacuum (unaffected by gravity or motion of medium).
19. **Astronomical Unit (AU)**: Average distance between center of Earth and center of Sun. $1\text{ AU} = 1.495978707 \times 10^{11}\text{ m} \approx 1.496 \times 10^{11}\text{ m} \approx 149.6\text{ million km}$.
20. **Parsec (Parallactic Second)**: Largest practical astronomical distance unit. Distance at which average radius of Earth's orbit (1 AU) subtends an angle of 1 arcsecond. $1\text{ Parsec} = 3.0857 \times 10^{16}\text{ m} = 3.262\text{ light-years} = 206,265\text{ AU} = \frac{648,000}{\pi}\text{ AU}$. Coined by Herbert Hall Turner (1913).
21. **Small Distance Units**:
22. **Height of a 6-foot tall person**: $1\text{ ft} = 0.3048\text{ m} \approx 30.5 \times 10^7\text{ nm} \implies 6\text{ ft} = 6 \times 30.5 \times 10^7\text{ nm} = 183 \times 10^7\text{ nm}$.
23. **Mass Units Conversion**: $1\text{ kg} = 10^3\text{ g}$; $1\text{ mg} = 10^{-3}\text{ g}$; $1\text{ }\mu g = 10^{-6}\text{ g}$; $1\text{ ng} = 10^{-9}\text{ g}$; $1\text{ pg (picogram)} = 10^{-12}\text{ g}$; $1\text{ fg (femtogram)} = 10^{-15}\text{ g}$.

24. **Pressure SI Unit**: **Pascal** (Pa). $1\text{ Pa} = 1\text{ N/m}^2 = 1\text{ kg}\cdot m^{-1}\cdot s^{-2}$.
25. **Atmospheric Pressure & Conversions**:
26. **Strain & Mach Number are Dimensionless**: Mechanical strain ($\Delta l / l$), Mach number (object speed / sound speed), Refractive index, Relative density, and Poisson's ratio have **NO units and NO dimensions**.
27. **Barrel of Oil**: $1\text{ barrel} = 158.9873\text{ litres} \approx 159\text{ litres} = 0.158987\text{ m}^3 = 42\text{ US gallons} = 34.9723\text{ UK gallons}$.
28. **Horsepower (hp)**: $1\text{ hp} = 746\text{ Watts}$. Metric horsepower $\approx 735.5\text{ W}$.
29. **Megawatt (MW)**: Unit of power generated at power stations. $1\text{ MW} = 10^6\text{ W} = 1,000\text{ kW}$.
30. **Kilowatt-hour (kWh)**: Unit of **electrical energy** (NOT power). $1\text{ kWh} = 3.6 \times 10^6\text{ Joules} = 1\text{ B.O.T. unit}$.
31. **Cusec (Cubic feet per second)**: Unit of rate of water flow in rivers/canals. $1\text{ cusec} = 1\text{ ft}^3/s = 28.317\text{ litres/second}$.
32. **NTU (Nephelometric Turbidity Units)**: Measures water turbidity (cloudiness/opaqueness due to suspended particulate matter).
33. **Solar Constant**: Solar radiant energy received per unit area at top of Earth's atmosphere at $1\text{ AU} \approx 1.4\text{ kW/m}^2$ (1361 W/m^2).
34. **Dobson Unit (DU)**: Measures atmospheric column ozone thickness. $1\text{ DU} = 0.01\text{ mm}$ thickness of pure ozone compressed to STP ($0^\circ\text{C}, 1\text{ atm}$). Represents $2.687 \times 10^{20}\text{ ozone molecules/m}^2$. Normal ozone layer $\approx 300\text{ DU}$; Ozone hole $<220\text{ DU}$.
35. **Nautical Mile & Knot**: $1\text{ Nautical Mile} = 1.852\text{ km} = 1852\text{ m}$ (distance at sea). **Knot** = speed of 1 nautical mile per hour ($1.852\text{ km/h} \approx 0.514\text{ m/s}$).

Chapter 11: Mechanics, Gravitation and Physical Properties of Matter

🎯 Consolidated Must-Score Facts

1. **Confirmed Moons Count in Solar System (2026 IAU Data)**:
2. **Saturn**: Maximum number of confirmed satellites in the solar system = **285 moons** (leading the solar system).
3. **Jupiter**: **101 confirmed moons** (Ganymede, Callisto, Io, Europa).
4. **Uranus**: 29 moons; **Neptune**: 16 moons; **Mars**: 2 moons (Phobos & Deimos); **Earth**: 1 moon; **Mercury & Venus**: 0 moons; Dwarf planet **Pluto**: 5 moons.
5. **71st BPSC (Pre 2025) Mechanics & Fluid Facts**:
6. **UKPCS (Pre 2025) Elasticity & Thermal Facts**:
7. **UP RO/ARO (Pre 2023) Moduli of Elasticity Facts**:
8. **70th BPSC (Pre 2024) Gas Liquefaction Condition**:
9. **Brahmagupta's Precedence**: 7th-century astronomer Brahmagupta (*Brahmasphuta Siddhanta*) stated that all bodies gravitate toward Earth long before Newton.
10. **Definition of Speed**: Galileo Galilei was the first to define speed mathematically as distance per unit time ($v = d/t$).
11. **Mass vs Weight**: Mass (kg, scalar, constant everywhere) measures inertia. Weight (N, vector, $W = mg$) depends on local gravitational acceleration.
12. **Deep Breath on Sensitive Scale**: Inhaling deeply increases scale reading because 4.8 L of inhaled air adds $\approx 5.88\text{ g}$ mass.
13. **Distance vs Displacement**: Distance is actual path length (scalar, ≥ 0). Displacement is shortest straight line (vector, can be zero).
14. **Average Speed for Equal Distance Halves**: Harmonic mean $v_{\text{avg}} = \frac{2 v_1 v_2}{v_1 + v_2}$.
15. **Newton's First Law (Inertia)**: Passenger leans backwards when vehicle starts (inertia of rest); leans forwards on sudden braking (inertia of motion); leans left when bus turns sharp right (inertia of direction).
16. **Newton's Second Law**: $F = \frac{dp}{dt} = ma$. Goalkeeper pulls hands backwards to prolong contact time (t), lowering rate of change of momentum and impact force.
17. **Newton's Third Law**: Action and reaction forces are equal, opposite, and act simultaneously on **two different bodies**.
18. **Propulsion by Whistling**: On frictionless ice, whistling/blowing air forward expels momentum, generating an equal backward reaction that propels the body (conservation of momentum).
19. **Boat Jump**: Jumping at 5 m/s from a boat that recoils at 0.5 m/s implies the boat is **10 times heavier** than the person ($M_{\text{boat}} = 10 M_{\text{man}}$).
20. **Friction Hierarchy**: **Static Friction $>$ Kinetic (Sliding) Friction $>$ Rolling Friction**.
21. **Loaded Cart Principle**: Harder to start moving than to keep rolling because limiting static friction exceeds dynamic kinetic friction.
22. **Ball Bearings**: Convert sliding/static friction into **rolling friction** by drastically minimizing contact surface area.
23. **Marginal Friction at Constant Speed**: $a = 0 \implies F_{\text{applied}} = f_{\text{friction}}$ ($200\text{ N} \implies 200\text{ N}$).
24. **Energy in Friction**: Moving against friction irreversibly dissipates **kinetic energy into heat energy**.
25. **Simple Machines**: Enable doing the **same amount of work with lesser effort force**; they never save total energy.
26. **Centripetal Force**: Inward radial force ($F_c = mv^2/r$) required for circular paths (gravity for orbits, tire friction for curves).
27. **Centrifugal Force**: Fictitious outward inertial force in rotating frames; drives cream separation from milk and washing machine spin drying.
28. **Overturning on Curves**: Unbanked vehicle moving too fast skids outwards and **overturns about its outer wheels**.
29. **Coriolis Force**: Due to Earth's rotation; deflects winds/currents to the **right in Northern Hemisphere** and **left in Southern Hemisphere**.

30. **Accelerated Liquid Tanker**: When a tanker accelerates forward, the liquid surface tilts backward, creating **larger depth at the rear end** ($\tan\theta = a/g$).
31. **Equal KE Stopping Distance**: Vehicles with equal KE stop over the **same distance ($X = Y = Z$)** under equal braking forces ($W = F \cdot d = KE$).
32. **de Broglie Wavelength**: $\lambda = h/p$. For 0.12 kg ball at 20 m/s, $\lambda = 2.76 \times 10^{-34}\text{ m}$.
33. **Energy Sources NOT from Sun**: **Geothermal energy** and **Nuclear energy** do NOT originate from solar radiation. Biomass and wind are solar-derived.
34. **Kepler's Laws**: 1st (elliptical orbits, Sun at focus); 2nd (areal velocity constant; fastest at perihelion, slowest at aphelion); 3rd ($T^2 \propto a^3$).
35. **Universal Gravitational Constant**: $G = 6.67430 \times 10^{-11}\text{ N}\cdot m^2/kg^2$ (measured by Henry Cavendish).

Chapter 12: Energy, Heat and Thermal Science

🎯 Consolidated Must-Score Facts

1. **70th BPSC (Pre 2024) Heat Transfer Without Medium Fact**:
2. **Thermal Radiation** is the only mode of heat transfer that does **NOT require any material medium** (propagates via electromagnetic waves across vacuum, e.g. solar energy reaching Earth). Conduction and convection strictly require material media.
3. **70th BPSC (Pre Re-Exam 2024) Steam Burn Severity Fact**:
4. **MP PCS (Pre 2023) Ecosystem Energy Degradation Fact**:
5. **69th BPSC (Pre 2023) Official Heat Wave Temperature Criteria**:
6. **67th BPSC (Pre 2022) Triple Point of Water Fact**:
7. **UPPCS (Pre 2025) Sweating & Humidity Fact**:
8. **UPPCS (Pre 2024) Humidity & Temperature Interdependence**:
9. **Relative humidity decreases** as air temperature rises (warm air has greater moisture-holding capacity). **Absolute humidity increases** with increasing surface evaporation.
10. **National Green Hydrogen Mission:** India targets **5 MMT (Million Metric Tonnes)** per annum green hydrogen production capacity by **2030**, supported by **125 GW** of associated renewable energy capacity. Reduces ₹1 lakh crore worth of fossil fuel imports. Hydrogen has the **highest calorific value** among all known chemical fuels (~150 kJ/g or 150 MJ/kg), producing only water vapor on combustion.
11. **PM Surya Ghar: Muft Bijli Yojana (February 2024):** Aims to install rooftop solar power systems in **1 crore (10 million) households**, providing up to **300 units of free electricity** per month with a total financial outlay of ₹75,021 crore.
12. **COP28 UAE Consensus & India's Target:** Tripling global renewable energy capacity by 2030. India's updated NDC (*Panchamrit* targets announced at Glasgow COP26) commits to achieving **500 GW of non-fossil energy capacity by 2030** and reducing emissions intensity of GDP by **45%** by 2030 relative to 2005 levels.
13. **India's First 100% Solar-Powered Airport: Cochin International Airport (CIAL)**, Kerala became the world's first fully solar-powered airport in August 2015, winning the UN Champions of the Earth Award.
14. **Ultra-Supercritical Thermal Power Units:** NTPC's deployment of ultra-supercritical coal units operating at pressures above 22.1 MPa and temperatures exceeding 600°C, elevating Carnot thermal efficiency to $>42\%$, thereby reducing specific coal consumption and CO_2 emissions.
15. **ITER & Tokamak Fusion Reactor Progress:** International Thermonuclear Experimental Reactor (Cadarache, Southern France). Magnetic confinement fusion utilizing deuterium-tritium plasma heated to $>150\text{ million }^\circ\text{C}$ (ten times hotter than the Sun's core). India is one of the seven key founding members providing the cryostat (built by L&T).
16. **Cryogenic Upper Stage (C32 / CE-20) in Gaganyaan & LVM3:** Liquid hydrogen fuel stored at -253°C (20 K) and liquid oxygen oxidizer stored at -183°C (90 K), delivering a very high specific impulse ($I_{sp} \approx 443\text{ s}$).
17. **Energy Conservation Building Code (ECBC) & Cool Roofs:** Implementing high Solar Reflectance Index (SRI) paints and phase-change building materials to block solar thermal infrared radiation, curbing indoor temperatures by $2\text{--}5^\circ\text{C}$.
18. **Work Formula & Sign:** Work done $W = \vec{F} \cdot \vec{d} = F d \cos\theta$. If force and displacement are perpendicular ($\theta = 90^\circ$, such as a porter carrying a load horizontally or centripetal force on a satellite), work done by that force is **identically zero**.
19. **Kinetic Energy & Momentum Relation:** $KE = \frac{p^2}{2m}$. If the momentum p of a body is doubled, its kinetic energy becomes **4 times**. If p increases by 100%, KE increases by 300%. If p increases by 50%, KE increases by 125%.
20. **Calorific Value Leader: Hydrogen** has the highest calorific value ($\approx 150\text{ kJ/g}$), followed by LPG ($\approx 50\text{ kJ/g}$), CNG/Methane ($\approx 50\text{ kJ/g}$), Petrol/Diesel ($\approx 45\text{ kJ/g}$), Coal ($25\text{--}33\text{ kJ/g}$), Dry wood ($17\text{ kJ/g}$), and Cow-dung cake ($6\text{--}8\text{ kJ/g}$).
21. **Mechanical Equivalent of Heat (J):** First determined by James Prescott Joule: $W = J \cdot Q$, where $J = 4.184\text{ J/cal} \approx 4.2\text{ J/cal}$. 1 calorie is the heat required to raise the temperature of 1 g of pure water from

14.5°C to 15.5°C at standard atmospheric pressure.

22. **Commercial Unit of Electrical Energy:** $1\text{ Board of Trade Unit (B.O.T.U.)} = 1\text{ kilowatt-hour (kWh)} = 1000\text{ W} \times 3600\text{ s} = 3.6 \times 10^6\text{ J} = 3.6\text{ MJ}$.
23. **Zeroth Law of Thermodynamics:** Formulated by Ralph H. Fowler (1931). States that if bodies A and B are each in thermal equilibrium with body C, then A and B are in thermal equilibrium with each other. This law introduces and **defines the concept of Temperature**.
24. **First Law of Thermodynamics:** $\Delta Q = \Delta U + \Delta W$. It is a direct manifestation of the universal **Law of Conservation of Energy**. It states that heat added to a system goes into increasing internal energy plus doing external work.
25. **Second Law of Thermodynamics:** Imposes direction on heat flow. **Clausius Statement:** Heat cannot spontaneously flow from a colder body to a hotter body without external work. **Kelvin-Planck Statement:** No engine can completely convert heat absorbed from a single reservoir into mechanical work in a cyclic process without rejecting heat.
26. **Concept of Entropy:** Clausius introduced Entropy (S, where $dS = dQ_{\text{rev}}/T$). In any irreversible natural process, the total entropy of the universe **always increases** ($\Delta S_{\text{univ}} > 0$).
27. **Absolute Zero Temperature:** The theoretical lowest possible temperature where molecular kinetic motion ceases and volume of an ideal gas drops to zero: $0\text{ K} = -273.15^\circ\text{C} = -459.67^\circ\text{F}$. Negative temperatures on the Kelvin scale are physically impossible.
28. **Triple Point of Water:** The unique thermodynamic point where solid ice, liquid water, and water vapor coexist in stable dynamic equilibrium: $T = 273.16\text{ K} = 0.01^\circ\text{C}$ at a vapor pressure of 611.65 Pa (4.58 mm of Hg).
29. **Anomalous Expansion of Water:** Pure water contracts upon heating from 0°C to 4°C, reaching its **maximum density ($1.000\text{ g/cm}^3 = 1000\text{ kg/m}^3$)** and **minimum volume** at **4°C (3.98°C)**. Above 4°C, it expands normally.
30. **Aquatic Survival in Frozen Lakes:** During severe winters, surface water cools to 4°C, becomes densest, and sinks to the lake bed. Once the entire lake reaches 4°C, surface water cools to 0°C, expands, becomes lighter, and freezes into floating surface ice. Because ice is a poor conductor of heat, the bottom water remains liquid at **$+4^\circ\text{C}$**, allowing aquatic fauna and flora to survive.
31. **Bursting of Water Pipes in Freezing Weather:** When liquid water freezes into ice at 0°C, its volume expands by approximately **9%**. In rigid metal or PVC pipes, the immense hydrostatic expansion pressure ruptures the pipes.
32. **Specific Heat Capacity of Water:** Exceptionally high: $c_{\text{water}} = 1\text{ cal/g}\cdot^\circ\text{C} = 4184\text{ J/kg}\cdot\text{K} \approx 4200\text{ J/kg}\cdot\text{K}$. It absorbs or releases large amounts of heat with minimal change in temperature.
33. **Why Water is Used as an Engine Coolant & in Radiators:** Because of water's high specific heat capacity, circulating water extracts massive heat energy from engine blocks without boiling rapidly.
34. **Why Water is Used in Hot Water Fermentation Bags:** Water holds thermal energy longer than any other common liquid due to its high specific heat capacity, providing prolonged, steady warmth.
35. **Maritime Climate Moderation:** Coastal areas (Mumbai, Chennai) experience mild winters and moderate summers because the adjoining sea acts as a giant thermal buffer due to water's high specific heat capacity. Conversely, continental interiors (Delhi, Nagpur) have extreme climates.

Chapter 13: Light, Optics and Laser Technology

🎯 Consolidated Must-Score Facts

1. **71st BPSC (Pre 2025) Optics Fact**:
2. **70th BPSC (Pre Re-Exam 2024) Dual Facts**:
3. **UKPCS (Pre 2024) Young's Double-Slit Numerical**:
4. **69th BPSC (Pre 2023) Colour Combination Fact**:
5. **68th BPSC (Pre 2022) Lens Intensity Fact**:
6. **Human Eye Physiological Response (UP RO/ARO 2021 Fact)**:
7. **Persistence of Vision** is **$1/16\text{th}$ of a second**; **Persistence of Hearing** is **$1/10\text{th}$ of a second**.
8. **Albedo & Fresh Snow (UPPCS Pre 2021)**:
9. **Luminous vs Non-Luminous Bodies**: A luminous object emits its own light (Sun, candle flame, electric lamp). The **Moon is non-luminous**; it is visible solely because it reflects incident sunlight.
10. **Speed of Light Hierarchy**: In vacuum, speed of light is maximum: $c = 3.0 \times 10^8\text{ m/s}$. In water: $2.25 \times 10^8\text{ m/s}$; in glass: $2.0 \times 10^8\text{ m/s}$; in diamond: $1.24 \times 10^8\text{ m/s}$. Speed is minimum in diamond (highest refractive index ≈ 2.42).
11. **Speed of Light in Water vs Vacuum**: $\frac{v_{\text{water}}}{c} = \frac{2.25 \times 10^8}{3.0 \times 10^8} = \mathbf{0.75}$ (three-fourths), NOT 0.95.
12. **Sunlight Travel Time**: Sunlight takes approximately **$499\text{ seconds} \approx 8.3\text{ minutes}$ (or 8.5 minutes)** to reach Earth across 1 AU.
13. **Invariance of Frequency in Refraction**: When light passes from one medium to another (e.g. air to glass), its **frequency (f) remains completely unchanged**, while its velocity (v) and wavelength (λ) change proportionally ($v = f \lambda$).
14. **Refraction Direction Rules**: Light entering an optically denser medium (higher refractive index) slows down and **bends towards the normal**. Light entering a rarer medium speeds up and **bends away from the normal**.
15. **Apparent Depth in Water**: A swimming pool or pond appears shallower than its actual depth due to refraction (light rays from bottom bend away from normal on exiting water). Real depth $d = \mu \times \text{apparent depth}$.
16. **Bent Stick & Raised Coin**: A pencil or stick partially immersed in water appears bent at the interface; a coin in a bucket appears raised due to refraction.
17. **Underwater Observer Looking Up**: An underwater diver looking at an aerial object sees the object at a **higher level than it actually is** (light travels from rarer air into denser water, bending toward the normal).
18. **Twinkling of Stars**: Caused by continuous **atmospheric refraction** through moving air layers of varying temperatures and optical densities.
19. **Early Sunrise and Delayed Sunset**: Due to atmospheric refraction, the Sun is visible **$\approx 2\text{ minutes}$ before actual geometric sunrise** and **$\approx 2\text{ minutes}$ after actual sunset**, lengthening the day by $\approx 4\text{ minutes}$.
20. **Elliptical Sun and Moon at Horizon**: Near the horizon, unequal atmospheric refraction across the upper and lower limbs flattens the vertical diameter, making them appear elliptical.
21. **Total Internal Reflection (TIR) Conditions**: (1) Light must travel from an optically **denser to a rarer medium**, and (2) Angle of incidence must exceed the **critical angle ($i > i_c$)**.
22. **Critical Angles for Common Media**: Water-air: $i_c \approx 48.75^\circ$; Glass-air: $i_c \approx 41.14^\circ$; Diamond-air: $i_c \approx \mathbf{24.4^\circ}$ (extremely small, causing repeated internal reflections).
23. **Sparkle of Diamond**: A cut diamond's brilliant sparkle is due to **Total Internal Reflection** enabled by its very high refractive index (2.42) and tiny critical angle (24.4°).
24. **Applications of TIR**:
25. **Soap Film & Bubble Colours**: The shimmering rainbow colors seen on soap bubbles and oil films in daylight are produced by the **Interference of light** reflecting from front and back surfaces.
26. **Compact Disc (CD) Iridescence**: A CD viewed in sunlight displays brilliant rainbow colors due to **Reflection and Diffraction** of light from microscopic pits and tracks behaving as a diffraction grating.

27. **Dispersion in a Prism**: White light splits into 7 constituent colors (VIBGYOR). **Violet light has the shortest wavelength, highest frequency, travels slowest in glass, has highest refractive index, and deviates the most**. **Red light has the longest wavelength, lowest frequency, travels fastest in glass, has lowest refractive index, and deviates the least**.
28. **Rainbow Formation Physics**: Rainbows are formed by the combined action of **refraction, dispersion, and total internal reflection** within falling raindrops.
29. **Rainbow Colors and Position**:
30. **Rayleigh's Law of Scattering**: The intensity of scattered light is inversely proportional to the fourth power of its wavelength: $I \propto \frac{1}{\lambda^4}$. Shorter wavelengths (blue/violet) scatter ≈ 16 times more than red.
31. **Blue Sky & Red Sunset**:
32. **Sky Appearance to Astronauts & High-Altitude Flyers**: At very high altitudes or in outer space where there is no atmosphere to scatter light, the sky appears **completely black**.
33. **Red Danger Signals**: Red light has the longest visible wavelength and suffers the **least atmospheric scattering**, allowing it to penetrate fog and smoke and remain visible from the greatest distance.
34. **Green Grass Appearance**: Grass appears green because its chlorophyll **reflects green light** while absorbing red, blue, and other wavelengths.
35. **Doppler Redshift in Astronomy**: Light from a receding celestial source or rocketship undergoes a Doppler shift towards longer wavelengths (redshift). A blue light emitted on a rocket moving away at $0.2c$ appears **yellow-orange** to an observer on Earth.

Chapter 14: Sound and Wave Motion

🎯 Consolidated Must-Score Facts

1. **UNEP 'Frontiers Report' on Noise Pollution (Moradabad Spotlight):**
2. **UPPCS Direct Question Fact:** WHO recommends ambient residential noise levels below **45 dB**. Any continuous exposure above **80 dB** constitutes hazardous noise pollution causing permanent auditory damage, hypertension, and sleep disorders.
3. **Insolation vs Terrestrial Radiation (IAS Pre 2024 / UPPCS 2022 Fact):**
4. **Albedo Measurements in Climate Science (UPPCS Pre Fact):**
5. **Fresh snow cover** exhibits the highest albedo on Earth ($80\% - 95\%$ reflecting power), causing skiers to suffer UV burns from reflected radiation. Open water exhibits the lowest albedo ($\sim 3\% - 5\%$).
6. **Telecommunication Frequency Bands & Space Waves (RAS Pre / UPPCS Fact):**
7. **Ultra High Frequency (UHF: $300\text{ MHz} - 3\text{ GHz}$)** cannot follow the curvature of Earth (ground waves) nor be reflected by the ionosphere (sky waves). They propagate via **space waves (line-of-sight propagation)**, which powers mobile telephony, Wi-Fi, and UHF television broadcasting.
8. **Acoustic Bathymetry & Deep Sea Submersibles (Samudrayaan / MATSYA 6000 Fact):**
9. **Gamma rays (γ)** have the shortest wavelength ($< 10^{-12}\text{ m}$), highest frequency, highest energy per photon ($E = h\nu$), and maximum penetrating power; used in radiotherapy to destroy cancerous cells.
10. **Wilhelm Conrad Röntgen** discovered X-rays ($1\text{ nm} - 10^{-3}\text{ nm}$); X-rays are not deflected by electric or magnetic fields because they carry no electric charge.
11. **Cosmic rays are NOT electromagnetic waves**; they are streams of extremely high-energy charged subatomic particles (chiefly protons and alpha particles) arriving from outer space.
12. **Sky wave propagation** depends on the reflection of medium- and shortwave radio waves from the electrically ionized gas layers of the **Ionosphere**.
13. **Edwin Howard Armstrong** invented wideband Frequency Modulation (FM) in 1935; FM operates at higher frequency ($88-108\text{ MHz}$) and wider bandwidth, offering complete immunity from amplitude noise and static.
14. **Infrasonic waves** have frequencies below 20 Hz; generated by earthquakes, volcanoes, ocean swells, avalanches, and elephants, whales, and rhinos.
15. **Ultrasonic waves** have frequencies exceeding $20,000\text{ Hz}$ (20 kHz); bats, dolphins, and porpoises use ultrasound for echolocation.
16. **Echocardiography (Echo test)** utilizes reflected ultrasound beams to produce dynamic real-time images of cardiac valves and chambers.
17. **Galton Whistle** emits ultrasonic frequencies ($> 20\text{ kHz}$), completely inaudible to human ears but heard clearly by dogs for training.
18. **Pitch** is governed strictly by **frequency** (and inversely by wavelength); high pitch produces a sharp, shrill voice (females, children), while low pitch produces a deep, grave voice.
19. **Loudness** is governed by the **amplitude** of wave vibration ($I \propto A^2$); perceived loudness is measured in decibels (dB).
20. **Timbre (Quality)** is governed by the waveform and harmonic overtones, allowing the human ear to distinguish between a Sitar and a Flute playing the identical musical note at identical loudness.
21. **Mach Number** is the ratio of object velocity to local speed of sound; Supersonic jets ($M > 1$) flying through the stratosphere emit nitrogen oxides (NO_x) that catalyze ozone layer depletion.
22. **Mechanical Waves:**
23. **Transverse Waves:** Particle motion is strictly orthogonal to wave propagation. Propagates via **shear stress** (crests and troughs). Can only propagate through media that possess shear modulus (solids, and specialized orbital motion on liquid surfaces). They **cannot propagate through the bulk interior of liquids or gases**.
24. **Longitudinal Waves:** Particle displacement is parallel to wave propagation direction, creating regions of compression (high density, high pressure) and rarefaction (low density, low pressure). Propagate via **bulk modulus**; can exist in solids, liquids, and

gases.

25. **Electromagnetic Waves:**
26. **Newton-Laplace Formula:**
27. **Determinants of Speed of Sound:**
28. **Effect of Temperature:** $v \propto \sqrt{T}$. Speed increases by approximately 0.61 m/s for every 1°C temperature rise in air.
29. **Effect of Humidity:** Moist air contains water vapour ($M = 18$), which is lighter than dry air ($M \approx 29$). Thus, humid air is less dense than dry air, and the speed of sound is **greater in humid air**.
30. **Effect of Pressure:** Pressure changes at constant temperature cause density to change in identical proportion ($P/\rho = \text{constant}$). Hence, **speed of sound is independent of pressure**.
31. **Effect of Medium State:** Rigidity/Elasticity determines speed: $v_{\text{solid}} > v_{\text{liquid}} > v_{\text{gas}}$.
32. **Pitch (Frequency):**
33. **Loudness (Amplitude):**
34. **Quality / Timbre:**
35. **Infrasound ($< 20\text{ Hz}$):** Generated before earthquake P-waves; perceived by elephants and whales for long-distance communication; stethoscope records cardiac infrasonic vibrations.

Chapter 15: Electricity, Magnetism and Electromagnetism

🎯 Consolidated Must-Score Facts

1. **Electric Vehicle (EV) Battery Cathode Chemistries (IAS Pre 2025 Fact):**
2. **Cathode (Positive Electrode):** Typically composed of **Cobalt, Lithium, and Nickel** (as NMC: LiNiMnCoO_2) or Lithium Iron Phosphate (LFP: LiFePO_4), coated on aluminium foil.
3. **Anode (Negative Electrode):** Constructed of **Graphite (Carbon)** coated on copper foil.
4. **Common Trap Fact:** Graphite is strictly an **ANODE** material, NOT a cathode component!
5. **Alternative Powertrain Vehicles (IAS Pre 2025 Fact):**
6. **Distributed Energy Resources (DER: IAS Pre 2024 Fact):**
7. **Pumped-Storage Hydropower (PSH) in Uttar Pradesh (IAS Pre 2024 Fact):**
8. **Silkyara Tunnel Lighting Calculation (UPPCS RO/ARO Pre 2023 Fact):**
9. **Wind Power Installed Capacity in India (CGPSC / MPPSC Fact):**
10. **Global Solar Silicon Wafer Production (IAS Pre Fact):**
11. **Current density ($\vec{j} = I/A$) is a VECTOR quantity** (pointing in the direction of positive charge motion), whereas electric current (I) and electric potential (V) are **scalars**.
12. **Nickel-Cadmium (Ni-Cd)** rechargeable batteries use nickel oxide hydroxide and metallic cadmium electrodes with potassium hydroxide (KOH) alkaline electrolyte; used in calculators, cordless tools, and transistors.
13. **Lightning** contains tremendous electrical energy ($10^8 - 10^9\text{ Joules}$), sufficient to vaporize moisture inside a tree and cause explosive combustion.
14. **CFL (Compact Fluorescent Lamp)** contains mercury vapour and argon; electric discharge excites mercury atoms to emit ultraviolet radiation, which strikes inner phosphor coating to fluoresce visible white light (22% energy efficiency; lasts $6,000 - 15,000\text{ hours}$).
15. **Light Emitting Diode (LED)** operates on the principle of **ELECTROLUMINESCENCE**; electrons recombine with electron holes in a direct-bandgap semiconductor (e.g. Gallium Arsenide/Phosphide), releasing energy directly as photons.
16. **Sodium vapour street lamps** are omnidirectional (360°) and emit monochromatic yellow light (589 nm), producing shadowy-black colour rendering compared to high-CRI multi-spectrum LEDs.
17. **Current and Current Density:**
18. **Ohm's Law & Resistance:**
19. **Power and Joule Heating:**
20. **Electric Fuse Wire:**
21. **Three-Pin Plug & Earthing:**
22. **Short Circuit Physics:**
23. **Faraday's Laws of Induction:**
24. **AC vs DC Machinery:**
25. **Direct current (DC) produces zero time-varying magnetic flux** ($d\Phi/dt = 0$), so transformers CANNOT function on DC. Connecting DC will overheat and burn the primary winding.
26. **Photovoltaic Solar Energy:**
27. **Distributed Energy Resources (DER):**
28. **Electric Vehicle Batteries:**
29. **The "Current Density is Scalar" Trap:**
30. **The "Transformer Works on DC" Trap:**
31. **The "Heavier Bulb has Less Resistance" Paradox:**
32. **The "Graphite in Cathode" Trap:**
33. **The "Fuse Wire has High Melting Point" Trap:**

Chapter 16: Electronics, Semiconductors and Computer Technology

🎯 Consolidated Must-Score Facts

1. **AI Action Summit, Paris (February 2025 - IAS Pre 2025 / 71st BPSC Fact):**
2. **Common Trap Fact:** The **USA and the UK DID NOT sign** this declaration!
3. **Quantum Computing Processors: Majorana 1 & Ocelot (IAS Pre 2025 Fact):**
4. **Majorana 1:** Unveiled by **Microsoft** on 19 February 2025, utilizing topological qubits aimed at achieving a million-qubit scale.
5. **Ocelot:** Quantum processor developed by **Amazon Web Services (AWS)**.
6. **National Supercomputing Mission (NSM) Milestones (November 2025 Status):**
7. **Fastest Supercomputer Installed in India: PowerEdge XE 9680** (installed at Shakti Cloud, Yotta Data Services, Mumbai; ranked 28th globally; $R_{\max} = 84.31\text{ Petaflops}$, manufactured by Dell, USA).
8. **Fastest Indigenous Supercomputer Developed in India: AIRAWAT-PSAI** (installed at C-DAC Pune; ranked 188th globally; $R_{\max} = 8.50\text{ Petaflops}$, $R_{\text{peak}} = 13.17\text{ Petaflops}$).
9. **World's Top 3 Supercomputers (Nov 2025):** 1. **El Capitan (USA)** (1.809 Exaflops), 2. **Frontier (USA)**, 3. **Aurora (USA)**.
10. **PARAM Series Deployment in India:**
11. **PARAM 8000:** India's first indigenous supercomputer, built by C-DAC Pune in 1991 under **Dr. Vijay Bhatkar** ("Father of Indian Supercomputers").
12. **PARAM Kamrupa:** IIT-Guwahati (inaugurated by President Droupadi Murmu, Oct 2022; 838 TFLOPS).
13. **PARAM Shivay:** IIT-BHU (first installed under NSM).
14. **PARAM Shakti:** IIT-Kharagpur.
15. **PARAM Pravega:** IISc Bengaluru (3.3 Petaflops).
16. **PARAM Ganga:** IIT-Roorkee.
17. **PARAM Ananta:** IIT-Gandhinagar.
18. **PARAM Sanganak:** IIT-Kanpur.
19. **PARAM Rudra:** High-performance systems at IUAC Delhi, IIT-Bombay, IIT-Madras, NCRA Pune, SNBNCBS Kolkata, IIT-Patna, and C-DAC Delhi.
20. **ANUPAM Series:** Supercomputer series developed indigenously by **Bhabha Atomic Research Centre (BARC)** for internal nuclear simulations (first was Anupam 860/4 in 1991).
21. **Historical Firsts in Indian Computing:**
22. **HEC-2M (Hollerith Electronic Computer-2M):** India's first digital computer, imported from England (designed by A.D. Booth) and installed in **1956 at the Indian Statistical Institute (ISI) in Calcutta**.
23. **TIFRAC (TIFR Automatic Calculator):** First digital computer built in India, commissioned in **1960 at TIFR Mumbai**.
24. **E-Governance Initiatives (2024–2025 Focus):**
25. **Suvidha 2.0 App:** Upgraded mobile application launched by the **Election Commission of India (ECI)** in October 2024 as a one-stop portal for candidates and political parties to track nominations, schedules, and campaign permissions.
26. **Gartner's 4-Phase Evolution Model of E-Governance (MPPSC Pre 2025 Fact):**
27. **Four Pillars of E-Governance: People, Process, Technology, Resources** (MPPSC Pre 2024).
28. **Generative AI & Enterprise Breakthroughs:**
29. **ChatGPT:** Developed by OpenAI, launched on **November 30, 2022**, based on the Generative Pre-trained Transformer Large Language Model architecture.
30. **Maharaja:** World's first airline Generative AI virtual agent deployed by **Air India** (2023), powered by Microsoft's Azure OpenAI service, handling over 1,300 customer query domains.
31. **FPGA (Field Programmable Gate Array):** Reconfigurable semiconductor ICs widely deployed for hardware acceleration in AI inference and edge computing (MPPSC Pre 2025).

32. **Bit (Binary Digit)** is the smallest unit of digital data (0 or 1); a **Byte** comprises **8 bits**, and a **Nibble** comprises **4 bits**.
33. **Charles Babbage** is celebrated as the "Father of Computing" for designing the **Analytical Engine (1834/1837)**, which pioneered the ALU, integrated memory, and flow control.
34. **Alan Turing** formulated the theoretical foundation of modern computer science and artificial intelligence (Turing Machine, 1936; Turing Test / Imitation Game, 1949).
35. **Primary (Internal) Memory** is semiconductor memory directly addressed by the CPU: volatile **RAM** (Random Access Memory) and non-volatile **ROM** (Read-Only Memory).

Chapter 17: Nuclear and Atomic Physics

🎯 Consolidated Must-Score Facts

1. **India's Three-Stage Nuclear Programme & Kalpakkam PFBR (High-Yield Science Fact):**
2. **Stage 1:** Pressurized Heavy Water Reactors (PHWR) fuelled by **Natural Uranium ($\text{U-238} + 0.7\%\text{ U-235}$)**, producing electricity and byproduct **Plutonium-239 (Pu-239)**.
3. **Stage 2:** Fast Breeder Reactors (FBR) fuelled by **Plutonium-239** with a Thorium-232 blanket, breeding fissile **Uranium-233 (U-233)**. Core loading commenced at the **500 MWe Prototype Fast Breeder Reactor (PFBR) at Kalpakkam, Tamil Nadu**.
4. **Stage 3:** Advanced Heavy Water Reactors (AHWR) fuelled by **Thorium-232 / Uranium-233**, unlocking energy independence.
5. **Monazite Placer Sands & Legal Monopoly in India (IAS Pre 2022 Fact):**
6. **Monazite** is a complex reddish-brown phosphate mineral of Thorium and rare-earth elements $[(\text{Ce, La, Nd, Th, Y})\text{PO}_4]$.
7. **Legal Fact:** Monazite is a prescribed strategic substance under the Atomic Energy Act, 1962. **Private entities are strictly prohibited from processing or exporting monazite**. Only government undertakings—**IREL (Indian Rare Earths Limited)** under DAE and **KMML (Kerala Minerals and Metals Ltd)**—are legally authorized to mine, process, and extract thorium compounds.
8. **Uranium Enrichment Levels for Civil vs Military Use (IAS Pre 2023 Fact):**
9. **Natural Uranium:** Contains only **0.7% U-235**; the remaining 99.3% is non-fissile U-238.
10. **Low Enriched Uranium (LEU):** Enriched up to **$3\% - 5\%$ U-235**, which is standard for commercial civil nuclear power reactors.
11. **Highly Enriched Uranium (HEU):** Enriched to $> 20\%$ U-235. **Weapons-grade uranium requires $\ge 90\%$ U-235** to achieve compact critical mass.
12. **Common Trap Fact:** Commercial electricity DOES NOT require 60% enriched uranium (IAS Pre 2023 statement debunked). India derives 43.5% of its electrical power from coal, not uranium.
13. **Narora Atomic Power Station (NAPS, Uttar Pradesh Fact):**
14. **International Non-Proliferation Treaties (NPT & CTBT Must-Score):**
15. **NPT (Non-Proliferation Treaty):** Opened for signature in 1968, in force 1970, extended indefinitely in 1995 (191 parties). Recognizes 5 Nuclear-Weapon States (tested before 1 Jan 1967: US, Russia, UK, France, China). **India, Pakistan, Israel, and South Sudan have never signed**. North Korea announced withdrawal in 2003.
16. **CTBT (Comprehensive Nuclear-Test-Ban Treaty):** Adopted 1996; requires ratification by all 44 Annex-2 states. Not in force because India, Pakistan, and North Korea have not signed, and US, China, Egypt, Iran, and Israel have not ratified. Russia revoked ratification in 2023.
17. **Nuclear fission** is the splitting of a heavy atomic nucleus (e.g. U-235, Pu-239) into two medium-sized fragments of comparable mass upon absorbing a thermal (slow) neutron.
18. **Critical Mass** is the minimum quantity of fissile material required to sustain a self-propagating nuclear chain reaction.
19. **Nuclear Reactor Fuel:** Fissile materials such as **Uranium-235 ($^{235}_{92}\text{U}$)**, **Plutonium-239 ($^{239}_{94}\text{Pu}$)**, or synthetic **Uranium-233 ($^{233}_{92}\text{U}$)**.
20. **Moderator:** Slows down energetic fast fission neutrons (2 MeV) to thermal speeds (0.025 eV) through elastic collisions so they can induce subsequent fission.
21. **Control Rods:** Absorb surplus slow-moving thermal neutrons to regulate or halt the fission rate. Inserting control rods dampens the chain reaction; withdrawing them increases reactivity.
22. **Cadmium is NOT a nuclear fuel**; it is strictly a neutron absorber used in control rods and radiation shielding.
23. **Coolant:** Extracts intense heat generated in the reactor core; typical coolants include light water, heavy water, liquid sodium metal (in fast breeder reactors), CO_2 gas, and helium.
24. **Nuclear Fusion:** The thermonuclear coalescence of two or more light nuclei at extreme velocity/temperature to form a heavier, more tightly bound nucleus, with a mass deficit converted to energy ($E = \Delta m c^2$).

25. **Mesons** were theoretically predicted in 1935 by Japanese physicist **Hideki Yukawa** as carriers of the strong nuclear force binding nucleons together.
26. **Transuranic elements** (elements with atomic number $Z > 92$, beyond Uranium) were synthesized primarily by **Glenn T. Seaborg** and his team at UC Berkeley (Neptunium, Plutonium, Americium, Curium), NOT by Enrico Fermi.
27. **Ernest Rutherford** famously expressed skepticism before the Royal Society that atomic energy would ever be harnessed industrially, misjudging the commercial viability of mass-energy conversion.
28. **Radioactivity** is an intrinsic nuclear property of unstable isotopes that undergo spontaneous decay by emitting alpha particles (^4_2He), beta particles (electrons/positrons), and gamma rays.
29. **Monazite sand** is an anhydrous phosphate of thorium and rare-earth elements, representing India's primary nuclear fuel resource ($13.15\text{ million tonnes}$).
30. **Thorium (Th-232)** is fertile, not fissile; it cannot sustain a chain reaction directly. It must capture a neutron in a breeder reactor to transmute into fissile **Uranium-233 ($^{233}_{92}\text{U}$)**.
31. **Advantages of Thorium over Uranium:**
32. **Enriched Uranium** refers to natural uranium that has undergone isotopic separation (via gas centrifuges or gaseous diffusion) to artificially elevate the concentration of fissile U-235 above its natural 0.7%.
33. **Kudankulam Nuclear Power Plant** utilizes Russian-designed **VVER-1000** light-water pressurized water reactors (PWRs), using enriched uranium and light water.
34. **Alpha decay (α):** Atomic number decreases by 2 ($Z \to Z - 2$) and mass number decreases by 4 ($A \to A - 4$).
35. **Beta-minus decay (β^-):** A nuclear neutron converts into a proton, emitting an electron and an antineutrino; atomic number increases by 1 ($Z \to Z + 1$), mass number A remains constant (produces **isobars**).

Chapter 18: Scientists, Discoveries, Defence & Space Technology, and Applications of Physics

🎯 Consolidated Must-Score Facts

1. **Mission Divyastra (Agni-V MIRV Test, 11 March 2024):**
2. **Agni Prime (Agni P) Milestones (2024–2025):**
3. **Pralay Quasi-Ballistic Missile Salvo Launch (31 December 2025):**
4. **RudraM-II Air-to-Surface Anti-Radiation Missile (29 May 2024):**
5. **BrahMos Supersonic Cruise Missile Export Milestones (2024–2026):**
6. **Phase-II Ballistic Missile Defence (BMD) System (24 July 2024):**
7. **Akashteer Air Defence Command & Control System (2024–2025):**
8. **VSHORADS (Very Short Range Air Defence System):**
9. **Scorpene Submarine Fleet Complete:**
10. **Multilateral Export Regimes & Arms Treaties:**
11. **New START Treaty:** Bilateral US–Russia treaty officially **expired on 5 February 2026** (Russia suspended participation in Feb 2023).
12. **NATO Expansion: Sweden officially joined NATO as its 32nd member on 7 March 2024** (following Finland as the 31st member in April 2023), ending over 200 years of military neutrality.
13. **HCoC (Hague Code of Conduct):** Reached **145 subscribing states** in 2024 with the entry of **Qatar** (India joined on 1 June 2016).
14. **Australia Group (43 members, India joined 2018)** and **Wassenaar Arrangement (42 states, India joined 2017)** export controls remain foundational.
15. **SpaDeX Mission (Jan–April 2025):** Autonomous space docking executed on **16 January 2025** (India 4th nation after US, Russia, China), undocked 13 March, redocked 20 April, electrical power transfer 21 April 2025. **POEM-4** safely de-orbited into the Indian Ocean on **4 April 2025** (Debris Free Space Mission - DFSM).
16. **Chandrayaan-4 (Cabinet Approval Sep 2024):** Rs. 2,104.06 cr lunar sample return mission (two LVM3 launches, lunar orbit docking, 36 months execution).
17. **Chandrayaan-5 / LUPEX (March 10, 2025):** ISRO Lander + JAXA Rover (Mitsubishi) exploring South Pole PSR water-ice deposits via H3-24L launch.
18. **Bharatiya Antariksh Station (BAS):** BAS-1 module launch by **2028**; operational station by **2035**; Indian crewed lunar landing by **2040**.
19. **NISAR:** Dual-frequency (L-band & S-band) SweepSAR radar launch on GSLV-F16 (**30 July 2025**).
20. **Aditya-L1:** Continuous solar observation from Halo orbit at Sun-Earth L1 (1.5M km); captured X6.3-class solar flare (Feb 2024); released 1st & 2nd global datasets (Jan & Feb 2025).
21. **MACE Cherenkov Telescope (Hanle, Ladakh):** Inaugurated **4 October 2024** by AEC Chairman Dr. Ajit Kumar Mohanty; world's highest ($4,270\text{ m}$) and Asia's largest imaging Cherenkov observatory, built by BARC with ECIL.
22. **Nobel Prizes in Physics (2023–2025):**
23. **2023:** Pierre Agostini, Ferenc Krausz, and Anne L'Huillier (attosecond light pulses).
24. **2024:** John J. Hopfield and Geoffrey E. Hinton (Artificial neural networks using statistical mechanics and spin-glass physics).
25. **2025:** John Clarke, Michel H. Devoret, and John M. Martinis (macroscopic quantum tunneling and energy quantization in electrical circuits).
26. **International Prize in Statistics:** C.R. Rao (2023); Grace Wahba (2025, 'mother of smoothing splines').
27. **Deep Ocean Mission & Samudrayaan Project (2024–2026):**
28. **CANI Submarine OFC Project:** Dedicated by PM Narendra Modi on **10 August 2020**, providing high-speed optical fibre connectivity between Chennai and Port Blair (plus 7 other islands) across 2,312 km of seabed.
29. **Indian Polar Research Stations & Observatories:**

30. **Antarctica:** Managed by National Centre for Polar and Ocean Research (**NCPOR, Vasco da Gama, Goa**). Stations: **Dakshin Gangotri** (1983–84; decommissioned and ice-submerged 1990), **Maitri** (1989, Schirmacher Oasis), and **Bharati** (commissioned 18 March 2012 at Larsemann Hills, Prydz Bay, constructed of 134 shipping containers).
31. **Arctic:** Research station **Himadri** established July 2008 at Ny-Ålesund, Svalbard, Norway. **IndARC** deployed on 23 July 2014 in Kongsfjorden fjord, Svalbard—India's 1st multi-sensor moored underwater observatory studying Arctic climate influence on the Indian Summer Monsoon.
32. **Polar Code:** IMO mandatory safety code for ships in polar waters entered into force on **1 January 2017** under SOLAS and MARPOL.
33. **Nanotechnology & Advanced Semiconductors (2024–2026):**
34. **IIT Roorkee Nanofabrication Facility (2025):** Uttarakhand's first semiconductor nanofabrication research facility established to accelerate India Semiconductor Mission, Nano Mission, and National Quantum Mission.
35. **ICAR-CIRCOT Nano-Cellulose Plant (Mumbai):** India's first pilot plant utilizing cotton linters and agro-waste to produce industrial-grade nano-cellulose.

Chapter 19: Chemistry Topic 1 — Atomic Structure and Periodic Classification

🎯 Consolidated Must-Score Facts

1. **IUPAC Superheavy Elements & Island of Stability:** Synthesizing superheavy transactinide elements in the 7th period: **Nihonium ($Z=113$, Nh)**, **Moscovium ($Z=115$, Mc)**, **Tennessine ($Z=117$, Ts)**, and **Oganesson ($Z=118$, Og)**. Research targeting $Z=119$ (Ununennium) and $Z=120$ in the hypothetical 8th period to confirm Glenn Seaborg's predicted 'Island of Stability' ($Z=114, N=184$).
2. **Rare Earth Elements (REEs) & Critical Minerals Mission:** India's Ministry of Mines identified **30 critical minerals** (including Lithium, Cobalt, Nickel, and the 17 Rare Earth Elements: 15 Lanthanides + Scandium + Yttrium). IREL operating monazite processing facilities in Odisha and Kerala; geological discoveries of Lithium reserves in Reasi (Jammu & Kashmir) and Degana (Nagaur, Rajasthan).
3. **Gallium and Germanium Global Export Controls:** Strategic semiconductor metalloids. Gallium ($Z=31$) and Germanium ($Z=32$) are critical for military radar electronics (Gallium Nitride, GaN), high-efficiency satellite photovoltaics, and fiber-optic infrared optics.
4. **National Quantum Mission & Trapped-Ion Atoms:** Development of neutral atom quantum processors utilizing Rydberg states of Group 1 alkali metals (**Rubidium-87, Cesium-133**) and alkaline earth ions (**Strontium-88, Ytterbium-171**).
5. **International Year of the Periodic Table (IYPT):** UNESCO commemorated the 150th anniversary of Dmitri Mendeleev's periodic system (1869).
6. **Subatomic Mass & Charge Hierarchy:**
7. **Electron:** Discovered by **J.J. Thomson (1897)** via cathode ray discharge tubes. Charge $= -1.602 \times 10^{-19}\text{ C}$; Mass $= 9.109 \times 10^{-31}\text{ kg} = 0.000548\text{ u}$ (approx. **$1/1836\text{th}$ of proton mass**). Charge-to-mass ratio (e/m) $= 1.7588 \times 10^{11}\text{ C/kg}$.
8. **Proton:** Observed as canal rays/anode rays by Eugen Goldstein (1886) and identified as the hydrogen nucleus by **Ernest Rutherford (1919)**. Charge $= +1.602 \times 10^{-19}\text{ C}$; Mass $= 1.6726 \times 10^{-27}\text{ kg} = 1.00727\text{ u}$.
9. **Neutron:** Discovered by **James Chadwick in 1932** by bombarding beryllium foils with alpha particles ($^9_4Be + ^4_2He \to ^{12}_6C + ^1_0n$). Charge $= 0$ (neutral); Mass $= 1.6749 \times 10^{-27}\text{ kg} = 1.00866\text{ u}$. **The neutron is the heaviest subatomic particle in an ordinary atom.**
10. **Rutherford's Gold Foil α-Scattering Experiment (1911):** Bombarded a thin gold foil (100 nm thick) with fast α-particles:
11. **Bohr's Quantum Postulates (1913):**
12. **Four Quantum Numbers:**
13. **Principal Quantum Number ($n = 1, 2, 3, \dots$ / K, L, M, N):** Dictates main energy level, electron shell, and average distance from nucleus. Max electrons in shell $= 2n^2$.
14. **Azimuthal / Orbital Angular Momentum ($l = 0\text{ to } n-1$):** Dictates orbital shape ($l=0 \to s$ spherical; $l=1 \to p$ dumbbell; $l=2 \to d$ double-dumbbell; $l=3 \to f$ complex).
15. **Magnetic Quantum Number ($m_l = -l\text{ to } +l$):** Dictates spatial orientation of orbitals in 3D space. Number of orbitals in a subshell $= 2l + 1$.
16. **Spin Quantum Number ($m_s = +1/2\text{ or } -1/2$):** Dictates intrinsic electron spin orientation (clockwise or counter-clockwise).
17. **Aufbau Principle:** Electrons occupy orbitals of the **lowest available energy level first** before occupying higher levels. Governed by the **$(n + l)$ rule**:
18. **Pauli's Exclusion Principle (1925): No two electrons in the same atom can possess the exact same set of all four quantum numbers.** Consequently, an orbital can hold a maximum of **two electrons**, and they must have **opposite (antiparallel) spins** ($\uparrow\downarrow$).
19. **Hund's Rule of Maximum Multiplicity:** Electron pairing in degenerate orbitals belonging to the same subshell (p, d, f) **cannot occur until each orbital is singly occupied with parallel spins** ($\uparrow$, $\uparrow$, $\uparrow$).
20. **Anomalous Electronic Configurations of Chromium and Copper:**

21. **Chromium ($Z=24$):** Expected $[Ar] 3d^4 4s^2$; Actual $= [Ar] 3d^5 4s^1$.
22. **Copper ($Z=29$):** Expected $[Ar] 3d^9 4s^2$; Actual $= [Ar] 3d^{10} 4s^1$.
23. **Isotopes:** Atoms of the same element having the **same atomic number (Z) but different mass numbers (A)**. Identical chemical properties, different nuclear stability.
24. **Isobars:** Atoms of different chemical elements having the **same mass number (A) but different atomic numbers (Z)** (e.g., $^{40}*{18}Ar$ and $^{40}*Ca$). Different chemical and physical properties.
25. **Isotones:** Atoms having the **exact same number of neutrons ($N = A - Z$)** but different atomic numbers (e.g., $^{14}_6C$ and $^{16}_8O$, both having 8 neutrons).
26. **Isoelectronic Species:** Atoms, molecules, or ions containing the **exact same total number of electrons** (e.g., $N^{3-}, O^{2-}, F^-, Ne, Na^+, Mg^{2+}, Al^{3+}$, each having 10 electrons).
27. **Dobereiner's Law of Triads (1829):** Arranged elements in groups of three with similar chemical properties. The atomic weight of the middle element is approximately the arithmetic mean of the other two:
28. **Newlands' Law of Octaves (1864):** English chemist John Newlands arranged elements in order of increasing atomic weight; every **eighth element showed properties similar to the first**, analogous to musical musical notes (Sa, Re, Ga, Ma...). **Valid only up to Calcium ($Z=20$)**.
29. **Mendeleev's Periodic Law (1869):** Russian chemist Dmitri Mendeleev stated: "*The physical and chemical properties of elements are periodic functions of their Atomic Weights.*"
30. **Modern Periodic Law (Henry Moseley, 1913):** English physicist Henry Moseley measured characteristic X-ray frequencies (ν) emitted by elements under electron bombardment, discovering $\sqrt{\nu} = a(Z - b)$. Proved that **Atomic Number (Z) is more fundamental than atomic weight**:
31. **Architecture of the Modern Long Form Periodic Table:** Based on Bohr's electronic configurations:
32. **18 Vertical Columns** called **Groups**.
33. **7 Horizontal Rows** called **Periods**.
34. **Period Breakdown:**
35. **Magic Numbers in the Periodic Table:** The electron capacities of successive shells that confer exceptional nuclear and electronic stability: **2, 8, 8, 18, 18, 32**.

Chapter 20: Chemistry Topic 2 — Matter, Solutions and Purification

🎯 Consolidated Must-Score Facts

1. **Matter** is defined as anything possessing mass, inertia, and occupying three-dimensional physical space.
2. **Plasma** (the fourth state) consists of superheated, highly ionized gas containing free electrons and positive ions; it conducts electricity, generates magnetic fields, and makes up $>99\%$ of the visible universe (stars, Sun, auroras, lightning, neon sign tubes).
3. **Bose-Einstein Condensate (BEC)** (the fifth state) forms when integer-spin boson gases of extremely low density (about one-hundred-thousandth the density of normal air) are cooled to ultra-low temperatures near absolute zero (0 K or -273.15°C).
4. **Phase Transitions**: Melting (Solid $\to$ Liquid), Freezing (Liquid $\to$ Solid), Vaporization/Boiling (Liquid $\to$ Gas), Condensation (Gas $\to$ Liquid), Sublimation (Solid $\to$ Gas directly), Deposition (Gas $\to$ Solid directly).
5. **Sublimation** occurs because the substance's vapor pressure exceeds atmospheric pressure before reaching its melting point. Examples: **Camphor, Naphthalene, Anthracene, Ammonium Chloride (NH_4Cl), Iodine (I_2), Dry Ice (solid CO_2)**.
6. **Triple Point of Water** is the unique thermodynamic coordinate of temperature and pressure at which solid ice, liquid water, and water vapor coexist in stable thermodynamic equilibrium: strictly 273.16 K (0.01°C) and 4.58 mmHg (611.65 Pa or 0.006 atm).
7. **Critical Temperature (T_c)** is the temperature above which a gas cannot be liquefied, regardless of the magnitude of pressure applied. For CO_2, $T_c = 31.1^\circ\text{C}$.
8. **Burning of a candle** is uniquely **both physical and chemical**: melting and vaporization of paraffin wax is a physical change, while the combustion of wax vapor into CO_2, carbon soot, and H_2O is a chemical change.
9. **Rusting of Iron**: When iron reacts with atmospheric oxygen and water vapor, hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$) is formed. Rusting is an oxidation chemical change that results in an **increase in the total mass of the iron object** due to the addition of bonded oxygen and water molecules.
10. **Solubility** of most solid solutes in liquid solvents increases with increasing temperature (endothermic dissolution), but decreases for exothermic dissolutions (e.g., $Ce_2(SO_4)_3$, CaO).
11. **Henry's Law** states that the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid ($m = k_H \cdot P$):
12. **Molarity (M)** = moles of solute per liter of solution (mol/L). Because liquid volume expands or contracts with temperature, Molarity is **temperature-dependent**.
13. **Molality (m)** = moles of solute per kilogram of pure solvent (mol/kg). Because mass is independent of temperature, Molality is **strictly temperature-independent** and preferred in high-precision thermodynamic and colligative calculations.
14. **Normality (N)** = gram equivalents of solute per liter of solution ($N = M \times \text{Valence factor}$ / n-factor).
15. **Parts per Million (ppm)** is used for ultra-trace concentrations (pollutants in air, fluoride/arsenic in drinking water): $\text{ppm} = \frac{\text{Mass of component}}{\text{Total mass}} \times 10^6$.
16. **Colligative Properties** are physical properties of dilute solutions that depend **strictly on the total number of solute particles** (ions or molecules) present, and are completely independent of the chemical nature, size, or identity of the solute.
17. **Relative Lowering of Vapor Pressure** (Raoult's Law)
18. **Elevation of Boiling Point** ($\Delta T_b = i \cdot K_b \cdot m$)
19. **Depression of Freezing Point** ($\Delta T_f = i \cdot K_f \cdot m$)
20. **Osmotic Pressure** ($\Pi = i \cdot C \cdot R \cdot T$)
21. **Automobile Antifreeze**: In sub-zero polar/winter conditions, **Ethylene glycol ($C_2H_6O_2$)** is added to vehicle radiator coolant water; it depresses the freezing point well below 0°C to prevent engine block cracking and elevates the boiling point to prevent summer overheating.
22. **De-icing of Frozen Highways**: Rock salt ($NaCl$) or calcium chloride ($CaCl_2$) is broadcast over icy roads. $CaCl_2$ is superior because it dissociates into 3 ions ($i=3$) and dissolves exothermically, melting snow down to -25°C.
23. **Pressure Cooker Mechanics**: By trapping steam, the internal chamber pressure rises to ~2 atmospheres (~15 psi gauge pressure), which **elevates the boiling point of water from 100°C to approximately 120°C–**

122°C. Food cooks substantially faster because it absorbs heat at a significantly higher temperature, not because of the pressure itself.

24. **Osmosis** is the spontaneous net movement of solvent molecules from a region of lower solute concentration (higher chemical potential of solvent) to higher solute concentration across a **semi-permeable membrane (SPM)**.
25. **Osmotic Pressure (Π)** is the exact excess hydrostatic pressure that must be applied to the solution side to completely halt the osmotic influx of pure solvent.
26. **Reverse Osmosis (RO)**: When an external hydrostatic pressure greater than the osmotic pressure ($P > \Pi$) is applied to the concentrated solution side, pure solvent molecules are forced backwards through the semi-permeable membrane into the pure water reservoir. This is the cornerstone technology for **desalination of seawater** and domestic water purifiers.
27. **Isotonic Solutions** possess identical osmotic pressure across a semi-permeable boundary. A $0.9\%\text{ (w/v) } NaCl$ solution ("normal saline") is strictly isotonic with human red blood cells (RBCs).
28. **Colloidal Solution**: A heterogeneous system wherein the dispersed phase particle diameter lies between **1 nm (10^{-9} m) and 1000 nm ($1\text{ }\mu\text{m}$ or 10^{-6} m)**. Colloidal particles pass through ordinary filter paper but are arrested by animal/parchment membranes.
29. **Tyndall Effect**: The scattering of a beam of visible light by colloidal particles suspended in a medium, illuminating the path of light (e.g., sunlight entering a dusty room through a slit, headlights in dense fog, blue appearance of the sky). True solutions do NOT exhibit the Tyndall effect.
30. **Brownian Motion**: The continuous, zig-zag, random kinetic bombardment of colloidal particles by solvent molecules (first observed by botanist Robert Brown in 1827). Brownian motion opposes gravity and confers physical stability upon colloidal sols.
31. **Electrophoresis (Cataphoresis)**: The migration of electrically charged colloidal particles toward the oppositely charged electrode under the influence of an applied external electric field; it definitively proves that **colloidal particles carry electric charge**.
32. **Coagulation (Flocculation)**: The process of neutralizing the electric charge of colloidal particles, causing them to aggregate into larger macro-particles and settle out as a precipitate.
33. **Hardy-Schulze Rule**: The coagulating (flocculating) power of an active electrolyte ion is directly proportional to the fourth power of its valence. For negatively charged colloidal particles (e.g., clay particles in muddy water, As_2S_3 sol), the coagulation power follows: **$Al^{3+} > Ba^{2+} > Na^+$**.
34. **Purification of Muddy Water by Potash Alum**: River and pond mud consists of negatively charged colloidal clay suspended in water. When potash alum ($KAl(SO_4)_2 \cdot 12H_2O$) dissolves, trivalent aluminium cations (Al^{3+}) neutralize the negative surface charges of clay particles, precipitating them rapidly through coagulation.
35. **Delta Formation**: River water carries negatively charged colloidal clay and silt. When river water enters the ocean, the abundant multivalent metal cations (Na^+, Mg^{2+}, Ca^{2+}) in seawater immediately coagulate and deposit the silt at the river mouth, forming triangular geological deltas.

Chapter 21: Chemistry Topic 3 — Metals, Non-Metals and Chemical Reactions

🎯 Consolidated Must-Score Facts

1. **Metals** are electropositive elements that readily donate valence electrons to form positive cations ($M \to M^{n+} + n e^-$).
2. **Gold (Au)** is the most **malleable** and most **ductile** metal known: a single gram of gold can be drawn into a microscopic wire over 2 kilometers long or beaten into gold leaf only 0.00001 mm thick.
3. **Silver (Ag)** has the **highest electrical and thermal conductivity** of all metals, followed in sequence by Copper (Cu), Gold (Au), and Aluminium (Al).
4. **Lead (Pb)** and **Mercury (Hg)** are the **poorest conductors of heat** among common commercial metals.
5. **Mercury (Hg, Hydrargyrum, $Z=80$)** is the only elemental metal that exists in the **liquid state** at standard room temperature (25°C); its freezing point is -38.83°C and normal boiling point is 356.7°C.
6. **Gallium (Ga, m.p. 29.76°C)** and **Caesium (Cs, m.p. 28.44°C)** are low-melting metals that liquefy directly in the palm of a human hand.
7. **Osmium (Os, density 22.59 g/cm^3)** is the **densest naturally occurring element**, closely followed by Iridium (Ir, 22.56 g/cm^3) and Platinum (Pt, 21.45 g/cm^3).
8. **Lithium (Li, density 0.534 g/cm^3)** is the **lightest metal** and lowest-density solid element; it floats easily on water and paraffin oil.
9. **Tungsten (W, Wolfram, $Z=74$)** has the **highest melting point of all metals** (3422°C); its high tensile strength and high electrical resistance make it the standard incandescent light bulb filament.
10. **Alkali metals (Sodium Na, Potassium K)** are exceptionally soft metals that can be **easily sliced with an ordinary table knife**; they have low densities, low melting points, and react vigorously with atmospheric moisture and oxygen, requiring storage submerged in **kerosene oil**.
11. **Phosphorus (P_4)** is an exceptionally reactive non-metal that spontaneously catches fire in moist air (yielding dense white fumes of P_4O_{10}); it is stored safely submerged under **water**.
12. **Bromine (Br_2, $Z=35$)** is the **only non-metal that is liquid** at room temperature; it is a dense, volatile, reddish-brown liquid with fuming, corrosive vapors.
13. **Iodine (I_2, $Z=53$)** is a non-metal that anomalously displays a **bright metallic-like luster** and sublimes into deep violet vapors upon gentle heating.
14. **Diamond** is an allotrope of carbon where each carbon atom is bonded tetrahedrally to 4 other carbons via rigid sp^3 covalent bonds; it is the **hardest known naturally occurring mineral** (10 on Mohs scale), an electrical insulator, but an exceptional conductor of heat (better than copper).
15. **Graphite** is an allotrope of carbon consisting of planar hexagonal sheets bonded via sp^2 hybridization with delocalized π electrons; it is soft, slippery (used as a solid dry lubricant), and **conducts electricity**.
16. **Fullerenes (C_{60}, Buckminsterfullerene)** consist of 60 carbon atoms arranged in a truncated icosahedron containing 20 hexagons and 12 pentagons, resembling a hollow soccer ball ("buckyball").
17. **Graphene** is a single two-dimensional planar sheet of carbon atoms packed in a honeycomb crystal lattice; it is nearly transparent, conducts electricity with zero effective mass electrons, and is >100 times stronger than structural steel.
18. **Amphoteric Oxides** react with both acids and strong bases to yield salt and water. The two classic examples are **Aluminium oxide (Al_2O_3)** and **Zinc oxide (ZnO)**:
19. **Neutral Oxides** exhibit neither acidic nor basic properties: Nitrous oxide (N_2O, laughing gas), Nitric oxide (NO), Carbon monoxide (CO), and Water (H_2O).
20. **The Reactivity Series of Metals** in descending order:
21. **Potassium (K) and Sodium (Na)** react violently with cold water, evolving hydrogen gas which catches fire spontaneously with an explosive, yellow/lilac flame.
22. **Calcium (Ca)** reacts moderately with cold water; the bubbles of liberated H_2 gas stick to the metal surface, causing the calcium lumps to **float to the top of the water**.

23. **Magnesium (Mg)** does not react with cold water; it reacts slowly with hot water and vigorously with steam to form magnesium oxide (MgO) and H_2.
24. **Aluminium (Al), Zinc (Zn), and Iron (Fe)** react neither with cold nor hot water, but react exclusively with **superheated steam**:
25. **Lead, Copper, Silver, Platinum, and Gold** do not react with water, hot water, or steam at any temperature.
26. **Nitric Acid (HNO_3)** is a powerful oxidizing agent. When metals react with HNO_3, hydrogen gas is generally NOT evolved; instead, HNO_3 oxidizes H_2 to H_2O and gets reduced to nitrogen oxides (NO_2, NO, N_2O).
27. **Aqua Regia ("Royal Water")** is a freshly prepared fuming mixture of **$3\text{ parts concentrated } HCl$ and $1\text{ part concentrated } HNO_3$** (strict volumetric ratio $3:1$). It dissolves noble metals like Gold (Au) and Platinum (Pt) by producing nascent chlorine and nitrosyl chloride ($NOCl$), forming soluble tetrachloroaurate complexes ($[AuCl_4]^-$).
28. **Bauxite ($Al_2O_3 \cdot 2H_2O$)** is the principal commercial ore of aluminium; it is purified by Bayer's chemical leaching and reduced electrolytically in the **Hall-Héroult process** using molten cryolite (Na_3AlF_6) and fluorspar (CaF_2) to lower the operating melting point from 2050°C to $\sim 950^\circ\text{C}$.
29. **Iron Ores**: Haematite (Fe_2O_3, red ore, $>65\% Fe$), Magnetite (Fe_3O_4, black ore, magnetic, up to $72\% Fe$), Limonite ($2Fe_2O_3 \cdot 3H_2O$), Siderite ($FeCO_3$).
30. **Iron Pyrites (FeS_2)** displays a brassy golden metallic luster and is popularly termed **"Fool's Gold"**; it is an ore of sulphur, not iron extraction.
31. **Cinnabar (HgS)** is the primary ore of Mercury; heating cinnabar in air roasts it directly into volatile mercury vapor, which condenses into liquid mercury ($HgS + O_2 \to Hg + SO_2$).
32. **Galena (PbS)** is the primary ore of Lead; **Pitchblende (U_3O_8)** is the primary ore of Uranium (from which Marie Curie discovered Radium and Polonium).
33. **Monazite sand**, abundant in the placer beach sands of Kerala, is the primary global resource for **Thorium (Th)** and rare earth elements (cerium, lanthanum).
34. **Froth Flotation Process** (invented by Mrs. Carrie Everson) is used selectively to concentrate **sulphide ores** (e.g., $ZnS, PbS, CuFeS_2$) using pine oil as a frothing collector and xanthates to make ore particles hydrophobic.
35. **Calcination vs. Roasting**: Calcination heats carbonate/hydrated ores in the **absence or limited supply of air** below melting point ($CaCO_3 \xrightarrow{\Delta} CaO + CO_2$). Roasting heats sulphide ores in a **regular, excess supply of air** ($2ZnS + 3O_2 \to 2ZnO + 2SO_2$).

Chapter 22: Chemistry Topic 4 — Acids, Bases, Salts and Carbohydrates

🎯 Consolidated Must-Score Facts

1. **Arrhenius Theory**: An **acid** is a chemical substance that dissociates in aqueous solution to yield hydrogen ions (H^+ or hydronium ions H_3O^+); a **base** dissociates to yield hydroxide ions (OH^-).
2. **Brønsted-Lowry Theory**: An **acid is a proton (H^+) donor**, and a **base is a proton (H^+) acceptor**. Every acid has a corresponding conjugate base, and every base has a conjugate acid.
3. **Lewis Theory**: An **acid is an electron-pair acceptor** (electrophile, e.g., $BF_3, AlCl_3, H^+$); a **base is an electron-pair donor** (nucleophile, e.g., NH_3, H_2O, OH^-).
4. **Acidic solutions** possess $[H^+] > 10^{-7}\text{ M}$, corresponding to **$\text{pH} < 7$**. The lower the pH value, the stronger the acidity.
5. **Basic (alkaline) solutions** possess $[H^+] < 10^{-7}\text{ M}$ and $[OH^-] > 10^{-7}\text{ M}$, corresponding to **$\text{pH} > 7$**. The higher the pH value, the stronger the alkalinity.
6. **Temperature Effect on pH**: Because auto-ionization of water ($H_2O \rightleftharpoons H^+ + OH^-$) is endothermic, increasing temperature shifts equilibrium to the right, increasing K_w. Consequently, at 50°C the pH of pure neutral water drops to ≈ 6.63, but the water **remains completely neutral** because $[H^+]$ remains exactly equal to $[OH^-]$.
7. **Human Blood pH**: Strictly maintained within the narrow, slightly alkaline window of **$7.35\text{ to }7.45$** (mean ≈ 7.40). A blood pH drop below 7.0 (acidosis) or rise above 7.8 (alkalosis) is fatal.
8. **Human Gastric Juice** contains hydrochloric acid (HCl, $\sim 0.5\%$), with a highly acidic pH of **$1.0\text{ to }1.5$**, which activates the protein-digesting enzyme pepsinogen into pepsin and sterilizes ingested pathogens.
9. **Acid Rain** is defined as precipitation with a **$\text{pH} < 5.6$**, caused primarily by atmospheric emissions of sulphur dioxide (SO_2) and nitrogen oxides (NO_x) reacting with rainwater to form sulphuric acid (H_2SO_4) and nitric acid (HNO_3).
10. **Natural Organic Acids & Food Sources**:
11. **Acetic Acid (Ethanoic Acid, CH_3COOH)**: Vinegar ($4–8\%$ aqueous solution).
12. **Formic Acid (Methanoic Acid, $HCOOH$)**: Stings of red ants, bees, wasps, and stinging nettle plant hairs.
13. **Citric Acid ($C_6H_8O_7$)**: Citrus fruits (lemons, oranges, limes, sweet limes).
14. **Tartaric Acid ($C_4H_6O_6$)**: Tamarind (imli), unripe grapes, cream of tartar; essential component of baking powder.
15. **Lactic Acid ($CH_3CH(OH)COOH$)**: Sour milk, curd (dahi), yogurt; muscle accumulation during anaerobic respiration causes muscular fatigue and cramps.
16. **Oxalic Acid ($H_2C_2O_4$)**: Tomatoes, spinach, sorrel; used industrially to remove stubborn rust/ink stains from textiles.
17. **Malic Acid ($C_4H_6O_5$)**: Apples (especially green/unripe apples), cherries.
18. **Ascorbic Acid (Vitamin C, $C_6H_8O_6$)**: Amla (Indian gooseberry), citrus fruits, guava.
19. **Tannic Acid**: Tea leaves (causes astringent taste).
20. **Butyric Acid**: Rancid, oxidized butter.
21. **Carbonic Acid (H_2CO_3)**: Carbonated soft drinks, club soda.
22. **Kidney Stones (Renal Calculi)**: Approximately 80% of all human kidney stones consist of insoluble crystals of **Calcium Oxalate (CaC_2O_4)**, formed when high concentrations of dietary oxalic acid bind with urinary calcium.
23. **Acid-Base Indicators**:
24. **Litmus**: Natural purple dye extracted from **Lichens** (Roccella tinctoria). Turns **Red in acidic solutions** and **Blue in basic/alkaline solutions**.
25. **Phenolphthalein**: Synthetic organic indicator. Remains **Colorless in acidic and neutral solutions**, but turns **Intense Magenta/Pink in basic solutions** (pH transition interval $8.2–10.0$).
26. **Methyl Orange**: Synthetic azo indicator. Turns **Red/Pink in acidic solutions** (pH < 3.1) and **Yellow in basic/neutral solutions** (pH > 4.4).
27. **Turmeric (Haldi)**: Natural yellow indicator. Remains yellow in acids/neutrals, but turns **Reddish-Brown in alkaline solutions** (explaining why yellow curry stains turn deep red when scrubbed with alkaline laundry soap).

28. **Red Cabbage Extract**: Natural anthocyanin indicator. Turns red/pink in acids, purple at neutral pH, and green-to-yellow in strong bases.
29. **Olfactory Indicators** are substances whose characteristic odor changes depending on whether they are in an acidic or basic medium. Examples: **Vanilla extract, Onion juice, and Clove oil** completely lose their smell in alkaline solutions ($NaOH$), but retain their scent in acids.
30. **Neutralization Reaction**: The exothermic chemical reaction between an acid and a base to produce a salt and water:
31. **Salt Hydrolysis and Solution pH**:
32. **Baking Soda**: Chemically **Sodium Hydrogen Carbonate / Sodium Bicarbonate ($NaHCO_3$)**:
33. **Baking Powder**: A dry commercial blend of **Baking Soda ($NaHCO_3$)** and a mild edible solid acid like **Tartaric Acid** or cream of tartar (potassium bitartrate).
34. **Washing Soda**: Chemically **Sodium Carbonate Decahydrate ($Na_2CO_3 \cdot 10H_2O$)**:
35. **Bleaching Powder**: Chemically **Calcium Oxychloride / Calcium Hypochlorite ($CaOCl_2$)**:

Chapter 23: Chemistry Topic 5 — Carbon, Organic Compounds, Polymers and Fibres

🎯 Consolidated Must-Score Facts

1. **Carbon ($Z=6$)** exhibits exceptional chemical versatility due to two foundational properties: **Catenation** (the unique ability of carbon atoms to link together via strong covalent bonds to form long open chains, branched structures, or rings) and **Tetravalency** (sharing four valence electrons to form 4 covalent bonds).
2. **Hybridization States of Carbon**:
3. **Alkanes** ($C_n H_{2n+2}$): Saturated hydrocarbons with single bonds. Formerly termed **paraffins** (Latin *parum affinis* = little affinity) due to their chemical inertness toward acids and bases under ambient conditions.
4. **Methane (CH_4)**: The simplest alkane; known historically as **Marsh Gas** because it is generated by anaerobic methanogenic archaebacteria in stagnant swamps, waterlogged paddy fields, ruminant animal digestive tracts (cattle cud-chewing), and decomposing municipal landfill sites. It is the primary constituent of Natural Gas ($\approx 85–95\%$) and Biogas/Gobar Gas ($\approx 55–70\%$).
5. **Alkenes** ($C_n H_{2n}$): Unsaturated hydrocarbons containing at least one carbon-carbon double bond ($C=C$). Known as **olefins** (oil-forming).
6. **Ethylene / Ethene (C_2H_4)**: The only gaseous **plant hormone** that naturally promotes fruit ripening and leaf abscission; also serves as the chemical monomer for manufacturing polyethylene/polythene.
7. **Alkynes** ($C_n H_{2n-2}$): Unsaturated hydrocarbons containing at least one carbon-carbon triple bond ($C \equiv C$).
8. **Acetylene / Ethyne (C_2H_2)**: Burns in pure oxygen to produce an intensely hot **Oxy-acetylene flame ($\approx 3200^\circ\text{C}$)** used for industrial cutting and welding of steel.
9. **Artificial Fruit Ripening**: Industrial ripening often uses **Calcium Carbide (CaC_2)**, which reacts with atmospheric moisture to release acetylene gas ($CaC_2 + 2H_2O \to Ca(OH)_2 + C_2H_2 \uparrow$). *Must-Score*: Commercial calcium carbide typically contains toxic trace impurities of arsenic hydride (arsine, AsH_3) and phosphorus hydride (phosphine, PH_3), leading FSSAI to ban its use in food trade.
10. **Benzene (C_6H_6)**: The archetypal aromatic hydrocarbon. Proposed by August Kekulé in 1865 as a planar hexagonal ring of 6 sp^2 carbon atoms with alternating double bonds. Thermodynamically stabilized by resonance with a delocalized cloud of 6π electrons complying with **Hückel's Rule ($4n+2$ π-electrons, where $n=1$)**.
11. **Allotropes of Carbon**: Pure elemental carbon exists as crystalline allotropes (**Diamond, Graphite, Fullerenes, Carbon Nanotubes, Graphene**) and amorphous forms (Coal, Charcoal, Lampblack/Carbon Black, Coke).
12. **Diamond**: Each carbon is sp^3 hybridized and bonded tetrahedrally to 4 other carbons in a rigid 3D framework ($C-C$ bond length $1.54\text{ \AA}$). It is the hardest known natural substance (Mohs 10), an electrical insulator, but an exceptional thermal conductor.
13. **Graphite**: Each carbon is sp^2 hybridized and bonded to 3 others in planar hexagonal rings ($C-C$ bond length $1.42\text{ \AA}$). Layers are held by weak van der Waals dispersion forces ($3.35\text{ \AA}$ gap). Delocalized π electrons allow it to **conduct electricity**. It is soft, slippery, and acts as a **solid dry lubricant**.
14. **Thermodynamic Stability of Carbon**: **Graphite is the thermodynamically most stable allotrope of carbon** under standard ambient conditions ($298.15\text{ K}, 1\text{ bar}$); its standard enthalpy of formation is defined as zero ($\Delta_f H^\circ = 0$). Diamond is metastable ($\Delta_f H^\circ = +1.9\text{ kJ/mol}$).
15. **Fullerenes (C_{60}, Buckminsterfullerene)**: Discovered in 1985 by Harold Kroto, Robert Curl, and Richard Smalley. Spherical cage resembling a soccer ball ("buckyball"), composed of **20 hexagonal rings and 12 pentagonal rings**.
16. **Carbon Nanotubes (CNTs)**: Discovered by Sumio Iijima in 1991. Cylindrical tubules of rolled graphene sheets with diameters on the nanometer scale; possess tensile strength >50 times that of steel and high thermal/electrical conductivity.
17. **Graphene**: Isolated in 2004 by Andre Geim and Konstantin Novoselov at Manchester University using adhesive tape exfoliation of graphite (Nobel Prize in Physics, 2010). A single atom-thick, two-dimensional honeycomb sheet of sp^2 carbon; nearly transparent (97.7% light transmission), ballistic electron conductor, and over 100 times stronger than structural steel.
18. **Black Gold**: Synthesized in 2019 by scientists at the Tata Institute of Fundamental Research (TIFR, Mumbai) by rearranging the inter-nanoparticle spacing of gold nanoparticles; absorbs visible and near-infrared solar light, acting as a solar water

desalination absorber and CO_2 conversion catalyst.

19. **Chronology of Landmark Advanced Materials**:
20. **Polymer**: A high-molecular-weight macromolecule formed by the repetitive chemical linkage of small, reactive monomer molecules ($M_w \approx 10^3\text{ to }10^7\text{ u}$).
21. **Addition (Chain-Growth) Polymers**: Formed by the repeated addition of unsaturated monomer units (containing $C=C$ or $C \equiv C$) without the elimination of any small byproduct molecules (e.g., Polyethylene, Polypropylene, PVC, Polystyrene, Teflon).
22. **Condensation (Step-Growth) Polymers**: Formed by repeated condensation reactions between bi-functional or poly-functional monomers, accompanied by the **elimination of small byproduct molecules** such as water (H_2O), alcohol (CH_3OH), or ammonia (NH_3) (e.g., Nylon-6,6, Terylene/Dacron, Bakelite, Melamine).
23. **Classification Based on Intermolecular Forces**:
24. **Elastomers**: Possess the **weakest intermolecular forces** (van der Waals forces). Polymer chains are randomly coiled and held together by occasional cross-links, allowing them to stretch under tension and snap back to original shape (e.g., Natural rubber, Vulcanized rubber, Buna-S, Buna-N, Neoprene).
25. **Fibres**: Possess the **strongest intermolecular forces** (extensive intermolecular hydrogen bonding or strong dipole-dipole interactions). Long, linear chains pack tightly into crystalline arrays, providing high tensile strength and high melting points (e.g., Nylon-6,6, Terylene, Kevlar, Silk, Wool).
26. **Thermoplastics**: Possess **intermediate intermolecular forces** between elastomers and fibres. Linear or slightly branched polymers that **soften on heating and harden upon cooling reversibly**; can be repeatedly melted, remoulded, and recycled (e.g., Polyethylene, PVC, Polystyrene, Teflon).
27. **Thermosetting Plastics**: Form extensive **three-dimensional cross-linked network polymers**. During moulding, heating triggers irreversible chemical cross-linking. Once set, they **cannot be remelted, softened, or reshaped**; excessive heating causes degradation. They are completely non-recyclable (e.g., Bakelite, Melamine-formaldehyde, Urea-formaldehyde).
28. **Polyethylene / Polythene**:
29. **Low-Density Polyethylene (LDPE)**: Polymerized under extreme pressure ($1000–2000\text{ atm}$) and temperature ($200–350^\circ\text{C}$) via a free-radical mechanism. Possesses extensive branched chains that prevent close packing, yielding low density and high flexibility; used for squeeze bottles, packaging films, and flexible tubing.
30. **High-Density Polyethylene (HDPE)**: Polymerized under low pressure ($6–7\text{ atm}$) and moderate temperature ($60–70^\circ\text{C}$) in the presence of a **Ziegler-Natta Catalyst** (triethylaluminium and titanium tetrachloride, $Al(C_2H_5)_3 + TiCl_4$). Consists of linear chains that pack tightly into crystalline structures, providing high density, rigidity, and chemical resistance; used for heavy buckets, dustbins, pipes, and milk crates.
31. **Polyvinyl Chloride (PVC)**: Formed by the addition polymerization of **Vinyl Chloride ($CH_2=CHCl$)**. An electrical insulator used for drainage pipes, electrical wire coatings, raincoats, and vinyl flooring.
32. **Teflon (PTFE, Polytetrafluoroethylene)**: Formed by the addition polymerization of **Tetrafluoroethylene ($CF_2=CF_2$)** under high pressure with persulphate catalyst:
33. **Bakelite (Phenol-Formaldehyde Resin)**: The first fully synthetic commercial plastic, invented in 1907 by Belgian-American chemist **Leo Baekeland**:
34. **Melamine-Formaldehyde Resin**: A thermosetting polymer formed by the condensation of Melamine (2,4,6-triamino-1,3,5-triazine) with Formaldehyde. Highly durable and impact-resistant; used to manufacture **unbreakable household crockery, dinnerware**, and decorative laminate countertops.
35. **Polystyrene**: Addition polymer of **Styrene (phenylethene, $C_6H_5-CH=CH_2$)**. In its foamed expanded form, it is commercially known as **Thermocol**; used as thermal insulation in refrigerators and packaging material for fragile electronics.

Chapter 24: Chemistry Topic 6 — Gases, Fuels, Energy and Combustion

🎯 Consolidated Must-Score Facts

1. **Calorific Value** is the total quantity of heat liberated by the complete combustion of a unit mass (1 kg) or unit volume (1 m^3) of a fuel, expressed in **kJ/kg** or kcal/kg.
2. **Hydrogen (H_2)** has the **highest calorific value of all known fuels ($150,000\text{ kJ/kg}$)**, roughly 3 times higher than petrol or diesel.
3. **Master Hierarchy of Calorific Values (kJ/kg)**:
4. **LPG (Liquefied Petroleum Gas)**:
5. **Ethyl Mercaptan (Ethanethiol, C_2H_5SH)**: Natural alkane gases (butane, propane, methane) are completely colorless and odorless. To detect accidental leaks instantly and prevent catastrophic household explosions, a minute concentration of foul, pungent **Ethyl Mercaptan** (resembling the odor of rotten cabbages or sulfur) is deliberately blended into commercial LPG cylinders.
6. **Natural Gas**: Naturally occurring gaseous fossil fuel extracted from underground geological reservoirs (often associated with crude petroleum deposits). Composed predominantly of **Methane (CH_4, $85–95\%$)**, with minor fractions of ethane (C_2H_6), propane, butane, and traces of nitrogen and CO_2.
7. **CNG (Compressed Natural Gas)**:
8. **LNG (Liquefied Natural Gas)**:
9. **Biogas (Gobar Gas)**:
10. **Water Gas (Synthesis Gas / Syngas)**:
11. **Coal Classification & Ranks** (by increasing geological age, carbon content, and calorific value):
12. **Peat**: Earliest geological stage of coal formation; brown fibrous mass containing **$<60\%$ Carbon**, high moisture ($>70\%$), burns with heavy smoke and low calorific value ($10,000–15,000\text{ kJ/kg}$).
13. **Lignite ("Brown Coal")**: Soft, crumbly coal containing **$60–70\%$ Carbon** and high residual moisture; lowest commercial rank of coal. In India, massive deposits are mined at **Neyveli (Tamil Nadu)** for thermal power generation.
14. **Bituminous Coal ("Soft Coal" / Commercial Coal)**: The most abundant and widely used commercial coal; black, dense, contains **$75–85\%$ Carbon** and low moisture; standard fuel for thermal electricity generation and metallurgical coking in blast furnaces.
15. **Anthracite Coal ("Hard Coal")**: The highest rank and geologically oldest coal; black, lustrous, contains **$>90–95\%$ Carbon**; burns slowly with a smokeless, pale blue flame, minimal ash, and the highest calorific value ($\approx 33,000\text{ kJ/kg}$). In India, found exclusively in small deposits in **Jammu and Kashmir (Kalakot, Reasi)**.
16. **Petroleum Fractions by Boiling Point**:
17. **Engine Knocking**: In an internal combustion engine, if the air-fuel mixture ignites prematurely (pre-ignition or detonation) before the spark plug fires, shock waves hit the cylinder walls, producing an audible metallic rattling noise known as **knocking**, which reduces power output and causes engine wear.
18. **Octane Rating (Octane Number)**:
19. **n-Heptane (C_7H_{16})** knocks violently and is assigned an **Octane Number = 0**.
20. **Iso-octane (2,2,4-trimethylpentane, C_8H_{18})** burns smoothly and is assigned an **Octane Number = 100**.
21. **Anti-Knock Additives**:
22. **Cetane Rating (Cetane Number)**:
23. **n-Hexadecane (Cetane, $C_{16}H_{34}$)** has a very short ignition delay and is assigned **Cetane Number = 100**.
24. **α-Methylnaphthalene** has a long ignition delay and is assigned **Cetane Number = 0**.
25. **Generations of Biofuels**:
26. **1G (First Generation)**: Produced from edible human food crops (sugarcane juice, corn starch, wheat, edible vegetable oils). Creates food-versus-fuel conflict.
27. **2G (Second Generation)**: Produced from non-food agricultural lignocellulosic residues (rice straw, wheat stalk, sugarcane bagasse, corn cobs, bamboo). India's commercial 2G bio-ethanol refinery is operational at **Panipat (Haryana)**.

28. **3G (Third Generation)**: Extracted from aquatic algal biomass (microalgae) grown in sewage or saline wastewater ponds; rapid growth rates, zero arable land needed.
29. **4G (Fourth Generation)**: Utilizes genetically modified photosynthetic microorganisms and solar fuels that capture and convert atmospheric CO_2 directly into biofuels.
30. **Ethanol Blending in India**:
31. **Ethanol (C_2H_5OH)**: Produced by yeast (*Saccharomyces cerevisiae*) fermentation of molasses (byproduct of sugar manufacturing), broken grains (damaged rice, maize), and agricultural cellulose.
32. **Blending Ratio**: E10 (10% ethanol, 90% petrol); E20 (20% ethanol, 80% petrol).
33. **Transesterification Reaction**: Vegetable oil triglycerides react with methanol (CH_3OH) in the presence of a base catalyst ($NaOH$ or KOH) to produce **Fatty Acid Methyl Esters (FAME, Biodiesel)** and a commercially valuable byproduct, **Glycerol (Glycerine)**.
34. **Combustion**: An exothermic redox chemical reaction between a fuel and an oxidant (usually atmospheric oxygen) accompanied by the evolution of heat and light.
35. **Carbon Monoxide (CO) Toxicity**: Known as the **"Silent Killer"**; a colorless, odorless, non-irritating toxic gas. When inhaled, CO binds to human hemoglobin with an affinity **$\approx 200–250\text{ times}$ greater than oxygen**, forming stable **Carboxyhemoglobin (CO-Hb$)**. This halts systemic oxygen transport to brain and heart tissues, causing headache, dizziness, unconsciousness, and asphyxiation within minutes (often occurring when charcoal braziers/angithis are burned inside closed, unventilated bedrooms during winter).

Chapter 25: Chemistry Topic 7 — Chemistry in Daily Life, Industry and Agriculture

🎯 Consolidated Must-Score Facts

1. **Soaps** are sodium or potassium salts of higher fatty acids (containing 12 to 18 carbon atoms) such as **Stearic acid ($C_{17}H_{35}COOH$)**, **Palmitic acid ($C_{15}H_{31}COOH$)**, and **Oleic acid ($C_{17}H_{33}COOH$)**.
2. **Saponification**: The chemical alkaline hydrolysis of animal fats or vegetable oils (triglycerides) with aqueous sodium hydroxide ($NaOH$) or potassium hydroxide (KOH), producing soap and a commercially valuable byproduct, **Glycerol (Glycerine, $CH_2OH-CHOH-CH_2OH$)**.
3. **Hard Soap vs. Soft Soap**: Soaps made with **Sodium hydroxide ($NaOH$)** are hard soaps (laundry washing bars); soaps made with **Potassium hydroxide (KOH)** are soft soaps (toilet luxury soaps, shaving creams, liquid handwashes).
4. **Salting Out of Soap**: Common salt (**$NaCl$**) is added to the saponification mixture to decrease the solubility product of soap, precipitating it out as a curdled solid through the common-ion effect.
5. **Micelle Cleansing Action**: A soap molecule consists of a long non-polar hydrophobic (water-repelling / oil-soluble) hydrocarbon tail and a polar hydrophilic (water-loving) ionic carboxylate head ($-COO^-Na^+$). In water, soap molecules assemble into spherical aggregates called **micelles**: the hydrophobic hydrocarbon tails point inward to trap non-polar oil/grease dirt droplets, while the hydrophilic polar heads face outward into water. Agitation forms an emulsion that is rinsed away.
6. **Why Soap Fails in Hard Water**: Hard water contains dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions. These cations react with soluble sodium soap molecules to precipitate an insoluble, sticky white curd called **Scum** (calcium/magnesium stearate):
7. **Synthetic Detergents (Syndets)**: Sodium salts of long-chain alkyl hydrogen sulphates or alkylbenzene sulphonic acids (e.g., **Sodium Lauryl Sulphate**, **Sodium Dodecylbenzene Sulphonate**).
8. **Why Detergents Work in Hard Water**: The calcium and magnesium salts of alkyl sulphates and sulphonates are **completely soluble in water**; hence, synthetic detergents form abundant rich lather in both soft and hard water without producing insoluble scum.
9. **Analgesics**: Pharmaceutical drugs that relieve or eliminate bodily pain without causing loss of consciousness. Classified into:
10. **Antipyretics**: Drugs used to reduce elevated body temperature in fever (e.g., **Paracetamol**, **Aspirin**, **Phenacetin**).
11. **Antiseptics**: Chemical antimicrobial agents applied to **living human and animal tissues** (wounds, cuts, skin ulcers, mouth infections) to kill or inhibit microbial growth without damaging tissue cells.
12. **Dettol Composition**: The commercial household antiseptic Dettol is a blend of **Chloroxylenol** ($\approx 4.8\%$) and **α-Terpineol** dissolved in pine oil and alcohol.
13. **Bithionol**: An organosulfur chlorinated compound added to toilet soaps to impart long-lasting deodorant and antiseptic properties.
14. **Tincture of Iodine**: An effective antiseptic solution consisting of **$2–3\%\text{ elemental Iodine } (I_2)$** dissolved in an alcohol-water mixture.
15. **Iodoform (CHI_3)**: A yellow crystalline solid with a medicinal odor, historically used as an antiseptic dusting powder for septic surgical wounds due to the slow liberation of free iodine.
16. **Boric Acid (H_3BO_3)**: A very mild, non-irritating antiseptic used as an ophthalmic soothing **eye-wash**.
17. **Disinfectants**: Chemical antimicrobial agents applied strictly to **non-living inanimate surfaces** (floors, toilets, drainage pipes, surgical instruments).
18. **The Phenol Concentration Rule**: A **0.2% aqueous solution of Phenol (C_6H_5OH) acts as an Antiseptic**, whereas a **1.0% aqueous solution of Phenol acts as a Disinfectant**!
19. **Chlorine Disinfection**: Chlorine gas at concentrations of **$0.2\text{ to }0.4\text{ parts per million (ppm)}$** is standardly used to disinfect municipal drinking water supplies and swimming pools.
20. **Antibiotics**: Chemical metabolic substances produced by microorganisms (fungi, bacteria) that inhibit the growth of or destroy pathogenic bacteria at low concentrations.

21. **Discovery of Penicillin**: The world's first true antibiotic, discovered serendipitously in **1928 by Sir Alexander Fleming** at St. Mary's Hospital, London, from the green mold **Penicillium notatum**. Purified as a clinical drug by Howard Florey and Ernst Chain (Nobel Prize in Medicine, 1945).
22. **Broad-Spectrum Antibiotics**: Effective against a wide range of both Gram-positive and Gram-negative bacteria (e.g., **Chloramphenicol**, **Ampicillin**, **Amoxicillin**, **Tetracycline**). Chloramphenicol is effective in treating typhoid, meningitis, and pneumonia.
23. **Antacids**: Weakly basic compounds administered orally to neutralize excess hydrochloric acid in gastric juice (HCl), elevating stomach pH above 3.5. Classic examples: **Milk of Magnesia ($Mg(OH)_2$)**, Aluminium Hydroxide gel ($Al(OH)_3$), and Sodium Bicarbonate ($NaHCO_3$). Modern antacids like **Ranitidine** and **Omeprazole** act by blocking histamine H_2-receptors or proton pumps (H^+/K^+ ATPase).
24. **Food Preservatives**: Chemical additives that prevent food spoilage caused by microbial fermentation, mold growth, or oxidative rancidity:
25. **Sodium Benzoate (C_6H_5COONa)**: The most widely utilized chemical preservative in commercial packaged foods (fruit squashes, tomato ketchup, jams, soft drinks). It metabolizes safely in the human liver into hippuric acid and is excreted in urine.
26. **Sodium Metabisulphite / Potassium Metabisulphite ($K_2S_2O_5$)**: Releases sulfur dioxide gas (SO_2), which inhibits bacterial and fungal enzymes in wines, fruit juices, and cut dried fruits.
27. **Table Salt ($NaCl$), Sugar Syrup, and Vinegar (CH_3COOH)**: Natural osmotic and acidic food preservatives used in traditional pickling and fruit preservation.
28. **Food Antioxidants**: Sacrificial reducing agents added to fats, oils, and fried snacks (potato chips) to retard oxidative rancidity: **BHA (Butylated Hydroxyanisole)** and **BHT (Butylated Hydroxytoluene)**. Flushing snack bags with **Nitrogen gas (N_2)** prevents oxygen from contacting edible oils.
29. **Glass**: An amorphous, non-crystalline, transparent supercooled inorganic liquid of high viscosity, manufactured by fusing silica (sand, SiO_2) with metal carbonates and cooling the melt rapidly without crystallization.
30. **Soda-Lime Glass (Soft Glass)**: Approximate composition $Na_2O \cdot CaO \cdot 6SiO_2$. Low melting point, inexpensive; used for window panes, beverage bottles, mirror plates, and cheap glassware.
31. **Potash Glass (Hard Glass)**: Made using potassium carbonate (K_2CO_3), calcium carbonate, and silica ($K_2O \cdot CaO \cdot 6SiO_2$). Highly resistant to chemical reagents and acids; used for high-temperature chemical laboratory equipment (flasks, combustion tubes).
32. **Pyrex Glass (Borosilicate Glass)**: Formed by substituting boron trioxide ($B_2O_3, \approx 13\%$) and alumina for alkali. Possesses an exceptionally **low thermal expansion coefficient** and high thermal shock resistance; does not crack upon rapid heating/cooling; used for standard laboratory glassware (beakers, burettes) and microwave oven cookware.
33. **Flint Glass (Optical / Lead Crystal Glass)**: Contains **Lead Oxide ($PbO, 20–30\%$)**. Possesses a high density, high refractive index, and high optical dispersion; used for manufacturing achromatic camera lenses, optical prisms, telescope objectives, and brilliant crystal chandeliers.
34. **Crookes Glass**: A specialty glass containing **Cerium Oxide (CeO_2)**. Cerium atoms possess electronic energy levels that selectively and sharply **absorb harmful ultraviolet (UV) rays** while transmitting visible light; used for high-protection sunglasses and optical UV shields.
35. **Photochromic Glass**: Specialty glass containing microscopic dispersed crystals of **Silver Chloride ($AgCl$) or Silver Bromide ($AgBr$)**. When exposed to sunlight (UV photons), silver halide decomposes into tiny clusters of metallic silver ($AgCl \xrightarrow{h\nu} Ag + Cl$), turning the lens dark. In dim indoor light, copper sensitizers catalyze recombination, restoring optical clarity.

Chapter 26: Chemistry Topic 8 — Nuclear Chemistry, Explosives and Environment

🎯 Consolidated Must-Score Facts

1. **Radioactivity** is the spontaneous, uncontrolled nuclear disintegration of unstable heavy atomic nuclei accompanied by the emission of ionizing particles or electromagnetic radiation.
2. **Nuclear Property**: Radioactivity is governed exclusively by nuclear stability (neutron-to-proton ratio, N/Z); it is completely **independent of temperature, pressure, chemical combination, or physical state**.
3. **Alpha (α) Particles**: Helium nuclei carrying a double positive charge (**$^4_2He^{2+}$**). Possess the highest mass (4 u), **highest ionizing power ($\approx 10,000\times \gamma$)**, but the **lowest penetrating power**; completely stopped by a single sheet of paper or the dead outer keratin layer of human skin.
4. **Beta (β^-) Particles**: Fast-moving electrons (**$^0_{-1}e$**) originating from the nucleus when a neutron decays into a proton, electron, and antineutrino ($n \to p + e^- + \bar{\nu}$). Moderate ionizing power and moderate penetrating power; stopped by a few millimeters of aluminium foil.
5. **Gamma (γ) Rays**: High-energy, short-wavelength electromagnetic photons emitted by excited nuclei transitioning to the ground state. Carry zero charge and zero rest mass; travel at the speed of light ($3 \times 10^8\text{ m/s}$). Possess the **lowest ionizing power**, but the **highest penetrating power** among nuclear emissions; require thick lead blocks or dense reinforced concrete barriers for shielding.
6. **Soddy-Fajans Group Displacement Law**:
7. **Radioactive Decay Kinetics & Half-Life ($t_{1/2}$)**:
8. **Units of Radioactivity**:
9. **Becquerel (Bq)**: SI unit; $1\text{ Bq} = 1\text{ nuclear disintegration per second}$.
10. **Curie (Ci)**: Traditional historical unit; $1\text{ Ci} = 3.7 \times 10^{10}\text{ disintegrations/sec}$ (the activity of 1 g of pure Radium-226).
11. **Rutherford (Rd)**: $1\text{ Rd} = 10^6\text{ disintegrations/sec}$.
12. **Radiometric Dating Techniques**:
13. **Radiocarbon Dating (^{14}C)**: Devised by Willard Libby (1949, Nobel Prize 1960). Based on the decay of cosmogenic ^{14}C ($t_{1/2} = 5,730\text{ years}$) into ^{14}N. Used for dating once-living biological archaeological artifacts (wood, bone, charcoal, fossils) up to $\approx 50,000\text{ years}$ old.
14. **Uranium-Lead Dating ($^{238}U \to {}^{206}Pb$)**: Half-life $\approx 4.5\text{ billion years}$; used to calculate the geological age of the Earth ($\approx 4.54\text{ billion years}$), ancient continental cratons, and lunar meteorites.
15. **Potassium-Argon Dating ($^{40}K \to {}^{40}Ar$)**: Half-life $\approx 1.25\text{ billion years}$; used to date ancient volcanic igneous rocks and early hominid fossil strata.
16. **Atomic Bomb (Fission Weapon)**:
17. **Nuclear Power Reactor (Controlled Fission)**:
18. **Moderator**: Material with light atomic mass used to slow down energetic fast neutrons (2 MeV) to thermal energy (0.025 eV) via elastic collisions without absorbing them. Standard moderators: **Heavy Water (D_2O, deuterium oxide)** and high-purity **Graphite**.
19. **Control Rods**: Neutron-absorbing materials inserted into the core to regulate reaction rate or shut down the reactor: **Cadmium (Cd)** and **Boron (B)** (have exceptionally high neutron capture cross-sections).
20. **Coolant**: Fluid circulated through the core to extract fission heat: **Light water (H_2O)**, Heavy water (D_2O), or liquid metallic Sodium (Na).
21. **Hydrogen Bomb (Thermonuclear Bomb)**:
22. **TNT (Trinitrotoluene)**:
23. **RDX (Research Department Explosive / Cyclonite / Hexogen / T4)**:
24. **PETN (Pentaerythritol Tetranitrate)**:
25. **ANFO (Ammonium Nitrate Fuel Oil)**:

26. **Gunpowder (Black Powder)**:
27. **Picric Acid (2,4,6-Trinitrophenol)**: A strong, yellow crystalline organic acid and explosive; historically used under the names **Lyddite** and **Melinite** as an artillery shell filling in the late 19th and early 20th centuries.
28. **Acid Rain (Acid Precipitation)**:
29. **Acid Rain Reactions**:
30. **Marble Cancer (Stone Leprosy)**:
31. **Ecological Damage of Acid Rain**:
32. **Classical Smog (London-type Smog)**:
33. **Photochemical Smog (Los Angeles-type Smog)**:
34. **Key Ingredients & Byproducts**:
35. **Ozone (O_3)**: Formed at ground level via photolysis of NO_2:

Chapter 27: Chemistry Topic 9 — Discoveries, Scientists and Miscellaneous Chemistry

🎯 Consolidated Must-Score Facts

1. **Antoine Laurent Lavoisier (1743–1794)**: Revered as the **"Father of Modern Chemistry"**. Formulated the fundamental **Law of Conservation of Mass** (1789), proved that combustion and respiration are chemical combinations with oxygen, disproved the ancient phlogiston theory, named the elements Oxygen and Hydrogen, and published the first modern list of chemical elements.
2. **John Dalton (1766–1844)**: Formulated the modern **Atomic Theory (1808)** (matter consists of indivisible atoms); established the **Law of Multiple Proportions** and **Dalton's Law of Partial Pressures**.
3. **Joseph Priestley (1774)** and **Carl Wilhelm Scheele (1772)**: Independently discovered **Oxygen gas (O_2)**. Priestley termed it "dephlogisticated air" by heating mercuric oxide; Lavoisier recognized it as an elemental gas and named it oxygen (meaning "acid-former").
4. **Henry Cavendish (1766)**: Discovered **Hydrogen gas (H_2)**, calling it "inflammable air" by reacting zinc with dilute sulphuric acid; later proved that burning hydrogen in air produces water, demonstrating that water is a compound, not an element.
5. **Humphry Davy (1778–1829)**: Pioneer of electrochemistry; used electric battery currents to isolate elemental **Sodium (Na), Potassium (K), Calcium (Ca), Magnesium (Mg), Strontium (Sr), and Barium (Ba)**; established that chlorine is an element; invented the **Davy Miner's Safety Lamp** to prevent firedamp explosions in coal mines; discovered the anesthetic properties of nitrous oxide.
6. **Friedrich Wöhler (1828)**: Synthesized **Urea (NH_2CONH_2)** by heating inorganic ammonium cyanate (NH_4CNO):
7. **Jöns Jacob Berzelius (1779–1848)**: Introduced the modern system of **chemical symbols using Latin letters** (H, O, C, Fe, Au); determined accurate atomic weights; discovered the elements Cerium, Selenium, Silicon, and Thorium; coined the terms *catalysis*, *isomerism*, *allotropy*, and *protein*.
8. **August Kekulé (1865)**: Proposed the cyclic hexagonal ring structure of **Benzene (C_6H_6)** with alternating double bonds, inspired by a famous daydream of a snake biting its own tail (Ouroboros); established the tetravalency of carbon.
9. **Harold Clayton Urey (1931)**: Discovered **Deuterium (^2H or D)**, the heavy isotope of hydrogen, by fractional distillation of liquid hydrogen (Nobel Prize in Chemistry, 1934).
10. **Linus Pauling (1901–1994)**: Formulated the quantum concepts of **orbital hybridization (sp, sp^2, sp^3)**, the nature of the chemical bond, the **Pauling Electronegativity Scale** (Fluorine = 4.0), and the α-helical secondary structure of proteins (Nobel Prize in Chemistry, 1954; Nobel Peace Prize, 1962).
11. **Acharya Prafulla Chandra Ray (P. C. Ray, 1861–1944)**:
12. **Sir Shanti Swaroop Bhatnagar (1894–1955)**:
13. **Chintamani Nagesa Ramachandra Rao (C. N. R. Rao, b. 1934)**:
14. **Asima Chatterjee (1917–2006)**:
15. **Water Chemistry — Hardness of Water**:
16. **Hard Water**: Water that does not produce lather easily with soap, but forms a sticky, insoluble white precipitate called **Scum** (calcium and magnesium stearates).
17. **Temporary Hardness (Carbonate Hardness)**:
18. **Boiling**: Converts soluble bicarbonates into insoluble carbonates/hydroxides that precipitate out:
19. **Clark's Method**: Adding a calculated quantity of **Slaked Lime ($Ca(OH)_2$)**:
20. **Permanent Hardness (Non-Carbonate Hardness)**:
21. **Washing Soda Method**: Adding calculated sodium carbonate (Na_2CO_3):
22. **Calgon's Method**: Adding **Sodium Hexametaphosphate ($Na_6P_6O_{18}$ or $(NaPO_3)_6$, Calgon = Calcium Gone)**:
23. **Permutit / Zeolite Process**: Passing hard water through a bed of hydrated sodium aluminium silicate (**Zeolite / Permutit**, $Na_2O \cdot Al_2O_3 \cdot xSiO_2 \cdot yH_2O$):
24. **Synthetic Ion-Exchange Resin Method**: Uses cation-exchange resins ($R-SO_3H$) to exchange Ca^{2+}, Mg^{2+} for H^+, followed by anion-exchange resins ($R-NH_3OH$) to exchange Cl^-, SO_4^{2-} for OH^-. Yields completely demineralized, deionized pure water.

25. **Heavy Water (D_2O, Deuterium Oxide)**:
26. **Heavy Water in Nuclear Power Plants**:
27. **Heavy Water Production in India**: Managed by the **Heavy Water Board (HWB)** under the Department of Atomic Energy (DAE). Major production facilities operate at **Manuguru (Telangana)** and **Thal (Maharashtra)** utilizing the Girdler-Sulphide (H_2S-H_2O) and ammonia-hydrogen exchange processes.
28. **Biological Toxicity of Heavy Water**: Ingesting high concentrations of D_2O is toxic to animals and plants. Deuterium forms stronger hydrogen bonds than protium, altering the 3D conformation of enzymes and significantly slowing metabolic biochemical reaction rates.
29. **Superheavy Water (T_2O, Tritium Oxide)**: Formed with radioactive Tritium (^3H); molecular mass $\approx 22\text{ g/mol}$. Tritium is a beta emitter with a half-life of 12.32 years.
30. **Hydrogen Peroxide (H_2O_2)**:
31. **Lunar Caustic**: Chemically **Silver Nitrate ($AgNO_3$)**.
32. **Red Lead (Sindoor / Minium)**: Chemically **Trilead Tetroxide (Pb_3O_4)**.
33. **Litharge**: Chemically **Lead Monoxide (PbO)**, yellowish-red powder used in lead glass and battery plates.
34. **Quicksilver**: Common historic name for elemental **Mercury (Hg)**, reflecting its fluid liquid mobility and bright silvery metallic luster.
35. **Vermilion (Cinnabar)**: Chemically **Mercuric Sulphide (HgS)**; deep crimson-red mineral pigment.